Resource-Driven Agricultural Clusters: Coagglomeration Effects of the High Plains Aquifer

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Abstract

Quantifying the regional economic impacts of crop production is critical for assessing the broader implications of access to natural resources and government policies. However, most existing estimates rely on input-output models with restrictive assumptions and competing theories of economic mechanisms provide little guidance on the sign or magnitude. We estimate local economic spillovers by exploiting a large increase in crop production that occurred due to the expansion of irrigation in counties overlying the High Plains Aquifer after World War II. While crop production was directly impacted by access to groundwater, we find that livestock product sales also increased significantly, reaching a 235% increase in sales compared to counties outside the aquifer. The primary spillover to non-farm sectors was an increase in sales in the wholesale sector, perhaps driven by sales of agricultural-related products. These findings provide evidence of coagglomeration incentives, where clusters of agricultural industries find it optimal to produce within the same region.

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1 Introduction

Increases in crop productivity result in national and global economic improvements (Gollin et al. 2002, 2021), but whether there are localized economic spillovers is unclear. Indeed, Allcott and Keniston (2018) indicate that economic spillovers observed at the cross-country level do not necessarily generalize down to subnational contexts. If crop production enhances local economic activity beyond crop production, then policymakers may support policies that enhance or preserve crop production, at least in part, due to these economic spillovers. On the other hand, if crop production has negative impacts on other sectors, then policymakers may seek to reallocate resources to offset the negative impacts. While understanding local economic spillovers is especially important for development policy, it is also relevant for assessing the broader impacts of productivity losses due to climate change or depleting resources.

While many studies suggest that crop production enhances broader local economic growth (Marbler 2024; Bustos et al. 2020; Lin et al. 2024), there are also many accounts of economic irrelevance, in which crop production does not interact with other industrial sectors. Indeed, Hornbeck and Keskin (2015) find that agricultural production is associated with the growth of non-agricultural sectors only in the short run and does not lead to long-run growth. Emerick (2018) finds similar patterns using district level data from India and Brown, Lambert, and Wojan (2019) rule out economically significant effects of converting cropland to conservation uses in the United States. More fundamentally, Foster and Rosenzweig (2004) conclude that "agricultural development and nonfarm activities are substitutes rather than complements" using village level data from India. Beyond agriculture, several papers study the impact of a resource discovery on local economic spillovers in the industrial and manufacturing sectors, but these studies yield divergent findings.[2]

There are several theoretical channels through which crop production may influence the local economy. On the positive side, it may increase local demand for goods and services by attracting population and raising incomes (Enrico 2011; Roback 1982; Rosen 1979). In addition, crop production could foster coagglomeration—the cross-industrial concentration of producers—through input-output linkages, shared labor markets, and knowledge spillovers (Ellison et al. 2010; Fujita et al. 2001; Marshall 1890). On the negative side, crop production may raise land and labor costs, discourage agglomeration—the concentration of firms within the same industry—among non-crop producers, and generate disamenities that reduce local demand (Corden and Neary 1982; Hornbeck and Keskin 2015). The net spillover effect depends on the relative sizes of the positive and negative effects.

In this paper, we estimate the local economic spillovers due to an increase in crop production from the expansion of irrigation in counties overlying the High Plains Aquifer in the Great Plains. The High Plains Aquifer was first discovered in the 1890s, but its extraction was limited until improved pumps and irrigation technology became widespread after World War II (Edwards and Smith 2018; Opie et al. 2018). Counties over the aquifer experienced plausibly exogenous gains in crop production when World War II technologies increased farmers' access to groundwater because the boundaries of the aquifer were determined by ancient geological processes not related to current agricultural productivity (Hornbeck and Keskin 2014, 2015). Our empirical model exploits this variation in groundwater accessibility for counties overlying the

[2] Allcott and Keniston (2018) and Kraus, Heilmayr and Koch (2023) find that resource sector has positive effect on manufacturing sector, while Harding and Venables (2016) find opposite results. Kline and Moretti (2014) finds manufacturing plants have positive effect on other plants.

aquifer after World War II compared to counties outside the aquifer using a difference-in-differences framework.

Our empirical model examines the effect of access to the aquifer on the livestock, wholesale, manufacturing, retail, and service sectors. We decompose the change in total production in these sectors to changes along the extensive and intensive margins.[3] One of the common identification concerns in local spillover research is the spatial equilibrium effect (Butts 2023; Clarke 2017). Crop production may also affect contiguous counties, but the treatment group is sharply defined by geographic boundaries. In this case, our estimation would violate the Stable Unit Treatment Value Assumption (SUTVA), so that the treatment effect may depend on the extent to which the spillovers propagate into the contiguous counties. Our model accounts for these spatial spillovers by not including counties immediately surrounding the aquifer in the control group (Butts 2023; Clarke 2017; Roth et al. 2023).

Key insights from our analysis are that livestock production and wholesale sales increased significantly over the past half century in the region due to the increase in crop production from the aquifer. Livestock sales increased substantially after 1960 and peaked at a 235% increase, primarily through increases in inventory and per-head value rather than the number of livestock farms. Such an increase in livestock sales is not an obvious theoretical prediction because corn can be readily traded across counties and local increases in land values and wages could crowd out livestock production. In the broader economy, we find that wholesalers' total sales, employment and revenue per establishment increase in the short and long run. We do not find a significant

[3] We define total margins as the changes in total production, the extensive margin as changes in the number of producers or establishments. For the livestock sector, the intensive margins include the value per head and the inventory of livestock per farm. For non-farm sectors, the intensive margins comprise changes in employment and output per establishment. This definition is analogous to Allcott and Keniston (2018).

effect on the number of wholesale establishments likely due to larger fixed and sunk costs than hiring workers. Meanwhile, we find only short-lived effects in retail and service sectors and no significant effects in manufacturing, which is less tied to local agricultural activity. Thus, our results suggest that crop production has minimal spillovers to sectors not within the agricultural supply chain.

Our work is most closely related to Hornbeck and Keskin (2014, 2015), but there are several important differences. First, while Hornbeck and Keskin (2015) focus on spillovers to non-farm sectors, we examine spillovers within the agricultural supply chain. Our results on the livestock sector indeed find evidence of significant coagglomeration incentives, which are important for policymakers. In dollar terms, we find that total livestock sales increased in counties overlying the aquifer by $23.6 billion in 1997 and $29.9 billion in 2017, both in 2022 dollars, representing a 235.2% and 196.9% increase in livestock product sales, respectively. While Hornbeck and Keskin (2015) conclude that "the agricultural sector did not encourage differential expansion of nonagricultural activity in counties over the Ogallala"—which our results also support—they ignore the large and important spillovers within the agricultural supply chain. Second, we provide information about the underlying spillover mechanisms by estimating different margins of adjustment as motivated by our conceptual framework. The evidence is more consistent with spillovers in the agricultural supply chain operating through input–output linkages that are central to coagglomeration, rather than with entry- or demand-driven mechanisms (Ellison et al. 2010). As a minor difference, we estimate the effects of access to the aquifer on wages across multiple non-farm sectors, whereas Hornbeck and Keskin (2015) focus only on the manufacturing sector. Because wage changes translate into income variations that ultimately shape local demand for goods and services, our broader coverage of sectoral wages provides a more complete picture.

This paper contributes to three areas of literature. The first area of literature that we contribute to examines how agricultural production affects economic growth by documenting the short- and long-run spillovers to livestock and wholesale sectors. Studies in this area leverage variety adoption,[4] water accessibility,[5] climate change,[6] or agricultural policies.[7] These studies focus primarily on the labor market and the manufacturing sector while none directly estimate spillovers in the livestock and wholesale sectors. Our results imply that input–output linkages play an important role in economic spillovers. Furthermore, we do not find short-run effects on cattle inventory but find large long-run effects, highlighting that the adjustments may occur over longer time horizons than those considered in much of the literature.

The second area of the literature that we contribute to explores the broader economic impacts of water availability. Input-output models such as IMPLAN are widely used to evaluate the economic effects of water availability (Dong et al. 2023; Guerrero et al. 2017; Howe et al. 1990; Howe and Goemans 2003), but input-output models cannot easily account for price adjustments, technological change, and scale economies, and are not likely to be suitable for long-term analysis (Loomis and Helfand 2003; Kowalewski 2009). We contribute by using an empirical

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[4] Within the literature utilizing variety adoption, Foster and Rosenzweig (2004, 2007) find that adopting high-yield varieties in India negatively affected the manufacturing sector. In contrast, Bustos, Caprettini and Ponticelli (2016, 2020) report that genetically engineered soybeans in Brazil generated positive spillovers on the manufacturing and service sectors.

[5] Studies exploiting variation in water access via canals (Blakeslee et al. 2023) and groundwater (Blakeslee et al. 2020; Dyer and Shapiro 2023; Hornbeck and Keskin 2015) find mixed impacts on manufacturing and employment.

[6] The literature on decreases in agricultural production due to climate shocks finds positive effects on non-agricultural employment and production in India (Colmer 2021; Emerick 2018; Liu et al. 2023), but also finds negative effects on broader economic activity using global grid-cell data (Marbler 2024).

[7] The literature that uses variations in policy also finds mixed results on the local labor market and manufacturing sector. Policies evaluated include cropland retirement in the U.S. (J. P. Brown et al. 2019; Sullivan et al. 2004), input subsidies in China (Mazur and Tetenyi 2024), export subsidies in Canada (Bollman and Ferguson 2019), trade liberalization in China (Lin et al. 2024), and production subsidies in Italy (Carillo 2021).

model that allows for economic adaptation. From another perspective, the water valuation literature emphasizes the importance of nonuse values within total economic valuation frameworks (C. E. Brown et al. 2025), but our findings suggest that even the use value may also be underestimated in the presence of economic spillovers. Our work also relates to research on the local economic impacts of water transfers (Ge et al. 2023; Howe et al. 1990; Howe and Goemans 2003) and large water infrastructure projects, including dams (Duflo and Pande 2007; Strobl and Strobl 2011).

More broadly, our paper contributes to a third area of the literature that examines resource curse and agglomeration economics. Studies using plausibly exogenous within-country shocks yield mixed results. For oil and gas booms, Allcott and Keniston (2018) and Caselli and Michaels (2013) find positive effects, while Arezki, Ramey and Sheng (2017) find no long-run impacts. Similarly, studies on mining booms produce divergent results across coal (Black et al. 2005; Glaeser et al. 2015), copper (Aroca 2001), and gold (Aragón and Rud 2013). We contribute to this literature by providing evidence from a natural experiment involving a water resource boom. Our findings show no evidence of a resource curse and instead suggest the presence of strong coagglomeration between sectors with closer input-output linkage.

2 Background on the High Plains Aquifer

This section provides background information on our study region and highlights how variation in accessibility to the aquifer supports our empirical analysis. We draw from various studies that provide an overview of crop production from the High Plains Aquifer (see also the discussion in Hornbeck and Keskin 2014). The High Plains Aquifer was first identified in 1898 (Darton 1903) and is one of the largest aquifers in the world, spanning approximately 174,000 square miles. The High Plains Aquifer is composed of several aquifers, but the Ogallala is the largest and most famous. Located in the central United States, east of the Rocky Mountains in the Great Plains, the aquifer underlies parts of Colorado, Kansas, Nebraska, New Mexico, Oklahoma, South Dakota, Texas, and Wyoming (Gutentag et al. 1984; McGuire 2017).

The boundaries of the High Plains Aquifer are defined by the location of ancient valleys and hills. These ancient landforms, shaped by erosion and sediment deposition millions of years ago, determined where permeable materials accumulated, allowing groundwater storage (Gutentag et al. 1984). As a result, variations in geological formations create differences in water availability across these boundaries, leading to distinct agricultural conditions on the modern-day land surface. This natural variation in groundwater access directly influences agricultural productivity, making the aquifer boundary a useful setting for studying economic differences (Perez-Quesada et al. 2024). The sharp delineation of the aquifer enables comparisons of economic outcomes between counties overlying the aquifer and their neighboring regions. While counties overlying the aquifer likely export or import goods to and from neighboring counties, we account for these spatial spillovers in our empirical model.

Extraction of the aquifer for irrigating crops remained limited before World War II because it was costly for farmers to irrigate using windmills, or steam and gasoline engine pumps. Even if windmill pumps reached the aquifer, they produced limited amounts of water and were used to furnish water for stock purposes.[8] In the 1930s, lessons from the Dust Bowl led some farmers to

[8] Hutson (1898) reported windmills in Texas could irrigate only about 7 acres per wheel, whereas pumping plants delivered enough water to support hundreds of acres.

adopt steam- and gasoline-powered pumps. These pumps could lift water f rom greater depths, but their high cost meant that most farmers viewed irrigation to be used only in emergency situations.[9] Due to these limitations, farmers overlying the aquifer in this period primarily relied on limited surface water diversions for irrigation (Evett et al. 2020).

Technological advancements and policy changes following World War II enabled farmers to extract groundwater from the High Plains Aquifer. Several factors played a significant role in the spreading of irrigation. First, automobile engines were adapted to power improved pumps after World War II, lifting groundwater cheaply and in larger volumes. Second, the introduction of aluminum and plastic pipes led to the expansion of closed conduit systems, which can reduce water loss compared to traditional open channels. In addition, the USDA and the Soil Conservation Service funded about one-third of the cost of installing a closed irrigation system (Opie et al. 2018). Third, since the 1950s, the adoption of electric pumps has increased with the expansion of the power grid, reducing reliance on traditional fuel sources. The post-World War II era saw a dramatic expansion of irrigation, which substantially contributed to the growth of crop production (Schoengold and Zilberman 2007).

The widespread adoption of center pivot irrigation technology beginning in the 1960s also brought about a rapid and substantial increase in crop production across the High Plains Aquifer region. Prior to this technological shift, farmers primarily relied on furrow or flood irrigation systems, which were ill-suited for the sandy soils and rolling terrain in some portions of the region (Opie et al. 2018). Moreover, conventional hand-move sprinkler systems were labor-intensive and uneconomical for most applications. In contrast, center pivot systems offered greater cost

[9] Opie, Miller and Archer (2018) report evidence of the high cost of installation and maintenance as a major barrier. Another challenge was the complexity of the early engines, which were difficult for farmers to operate. As a result, the lack of technical knowledge hindered widespread adoption.

efficiency, reduced labor demands, and the ability to operate effectively on uneven terrain (Lichtenberg 1989). These advantages contributed to a dramatic expansion in irrigated agriculture (McGuire 2017).

This shift created a stark difference in access to groundwater between farms located over the aquifer and those outside it. Farms overlying the aquifer could expand irrigation and increase productivity, whereas those outside the aquifer remained dependent on rainfall and limited surface water supplies. We leverage this historical variation in access to groundwater from the High Plains Aquifer to analyze the local economic spillovers of crop production. By leveraging the boundary of the aquifer, we examine how differences in crop production, driven by groundwater availability, affected production in various sectors before and after World War II. We note that when farmers withdraw water from the aquifer for irrigation, the saturated thickness decreases. This reduction in saturated thickness can eventually deplete some areas' potential for large-scale irrigation. In other words, water withdrawals influence saturated thickness, so changes in saturated thickness may be endogenous with economic activity. Our empirical analysis focuses on access to the High Plains Aquifer defined prior to extraction as a source of variation rather than relying on water levels.

Building on this setting, the next sections present our conceptual framework, data sources and empirical strategy to identify the local spillover effects of crop production on local economic outcomes.

3 Conceptual Framework

3.1 Conceptual Model

This section is designed to provide conceptual background for our empirical analysis. While we use plausibly exogenous historical variation in crop production, it is essential to understand the economic mechanisms through which crop production generates either positive or negative spillovers to non-crop sectors. Our simple conceptual model builds on the insights from Greenstone et al (2010), Moretti (2011), and Allcott and Keniston (2018).

We consider two non-crop production sectors, indexed by $j \in \{m, l\}$, representing tradable and local goods, and two counties, indexed by $c \in \{a, b\}$. County a overlies the High Plains Aquifer, whereas county b lies outside the aquifer. We suppose that the two counties are symmetric at $t = 0$, but crop production increases only in county a for $t > 0$ due to irrigation.[10] A producer in non-crop sector j and county c chooses aggregate inputs N_{jct}, such as labor and land, to maximize profits $P_{\mathrm{jct}}\,\Theta_{\mathrm{jct}}\,N_{\mathrm{jct}}^{1-\gamma} - W_{\mathrm{jct}}N_{\mathrm{jct}}$, where P_{jct} denotes output price, Θ_{jct} denotes productivity, W_{jct} denotes the cost of inputs, and $\gamma \in (0,1)$.

Productivity Θ_{jct} has several different components. Using lowercase letters to denote natural logs, we specify the log productivity as

$$\ln(\Theta_{\mathrm{jct}}) = \theta_{jct} = \rho_{jt} + \rho_{jc} + \Lambda_j f_{jct} + \varsigma_j v_{jct}, \qquad (1)$$

[10] The symmetry assumption is not essential but simplifies the exposition. See the supplementary appendix for the more general expression.

where ρ_{jt} denotes sector-specific productivity that varies over time and ρ_{jc} denotes time-invariant sector-by-county productivity. Agglomeration effects on productivity are modeled as the log of the number of producers in industry j (f_{jct}) times a productivity effect Λ_j. Coagglomeration effects on productivity are modeled as an index of crop production relevant for sector j (v_{jct}) times a productivity effect ς_j.

We model agglomeration spillovers through the number of firms in sector j. Definitions of agglomeration vary across studies. Some studies broadly define it as productivity improvements due to more firms within the region (Bleakley and Lin 2012; Allcott and Keniston 2018), but others narrowly define it as productivity improvements due to more firms within the same industry in a given region (Serafinelli 2019; Kerr and Kominers 2015; Rosenthal and Strange 2001, 2003). We adopt the narrower definition to distinguish within-sector agglomeration from cross-sector coagglomeration. A large body of literature empirically documents the presence of positive agglomeration spillovers with the narrower within-industry definition (Serafinelli 2019; Rosenthal and Strange 2001, 2003). Although the exact specification of agglomeration varies across studies, our specification is consistent with Kerr and Kominers (2015), Serafinelli (2019), Greenstone (2010), and Rosenthal and Strange (2001).

The coagglomeration spillover ($\varsigma_j \geq 0$) depends on how strongly sector j is linked to the crop sector. The coagglomeration literature develops indices to measure the strength of coagglomeration (Ellison et al. 2010; Howard 2015; O'Sullivan and Strange 2018), typically based on pairwise colocation patterns, and relates these indices to input-output linkages and other sources of interindustry spillovers, so our conceptualization of coagglomeration is also in line with the coagglomeration literature. Kraus et al. (2023), Emerick (2018), and Ellison et al. (2010) provide empirical evidence of coagglomeration spillovers.

After deriving indirect profits[11], taking log differences across counties a and b (Δ^c) and then differencing again between period 0 and period $t > 0$ gives the effect of crop production due to access to irrigation on sector j:

$$\Delta^c \pi_{jt} - \Delta^c \pi_{j0} = \underbrace{\frac{1}{\gamma}\Delta^c p_{jt}}_{\substack{local\ demand\\ effect}} + \underbrace{\frac{\Lambda_j}{\gamma}\Delta^c f_{jt}}_{\substack{agglomeration\\ effect}} + \underbrace{\frac{\varsigma_j}{\gamma}\Delta^c v_{jt}}_{\substack{coagglomeration\\ effect}} \underbrace{- \frac{1-\gamma}{\gamma}\Delta^c w_{jt}}_{\substack{input\ cost\\ effect}}. \quad (2)$$

Crop production can affect profits in other sectors through several mechanisms: through changes in output prices $\Delta^c p_{jt}$, the number of firms $\Delta^c f_{jt}$, coagglomeration $\Delta^c v_{jt}$, and input costs $\Delta^c w_{jt}$.

The local demand effect in Equation (2) captures that higher crop production could shift the demand for local goods and services upward by increasing local income and population, causing the price of local goods to increase ($\Delta^c p_{lt} \geq 0$).[12] In contrast, the price of tradable goods does not change because it is determined in national or global markets ($\Delta^c p_{mt} = 0$). Therefore, the local demand effect is expected to be nonnegative.

The input cost effect in Equation (2) captures that increases in crop production may raise input costs by reallocating local inputs, such as land and labor, toward the crop sector $\left(\Delta^c w_{jt} \geq 0 \right)$.[13] The input cost effect is expected to be nonpositive.

[11] See the supplementary appendix for a derivation.

[12] See conceptual model in Enrico (2011) and empirical results in Hornbeck and Keskin (2014, 2015), Allcott and Keniston (2018), and Bleakley and Lin (2012).

[13] See Sachs and Warner (1999, 2001). Resource reallocation and higher input costs are key channels of the resource curse.

The agglomeration effect captures that a change in the number of firms in sector j affects their profits through the agglomeration spillover coefficient. A Roback-Rosen style spatial equilibrium model indicates that changes in expected profits create profit differentials between counties a and b, which encourages firms to locate in the more profitable county (Roback 1982; Rosen 1979). However, the sign of the agglomeration effect is ambiguous because it depends on the relative strength of the local demand and input cost effects, as well as idiosyncratic firm preferences.

The coagglomeration effect captures that crop production could benefit non-crop sectors through input-output linkages, labor market pooling, and knowledge spillovers (Marshall, 1890; Fujita et al., 2001). Incentives for input-output linkages arise when industries supplying inputs for crop production or using the outputs find it beneficial to locate within the same region. Industries could also benefit from a larger pool of labor available locally. Knowledge spillovers occur when expertise gained in one industry extends to other industries. Thus, the coagglomeration effect is expected to be nonnegative.

We categorize a change in the cost of crops used as an input in non-crop sectors as a coagglomeration effect since it reflects an input-output linkage—consistent with previous literature. More specifically, in the corn-livestock setting, we consider the feed procurement advantages as a major component in the coagglomeration effect. Larger corn production can lower the effective cost of feed procurement for livestock producers by reducing transportation, storage, contracting, and search costs. The importance of lower transportation costs for feed is highlighted by the fact that feed represents roughly 67-77% of the cost of gain in beef cattle (Gustafson and Van Arsdall 1970). An alternative modeling approach would incorporate the crop sector as a separate input supply or output demand, but this would unnecessarily complicate the stylized conceptual model without any key unique insight.

Taken together, the local economic spillovers of crop production are theoretically mixed. We also recognize that an additional effect not reflected in our conceptual model above is that some crop practices may cause disamenities that drive population outflows, suppress local economic demand, and bid into higher wages (Hornbeck and Keskin 2015). The net effect of the local economic spillovers of crop production depends on the relative extent of the positive and negative effects. Identifying these effects empirically is essential to understanding how crop production affects local economies. Although our conceptual model reflects how profits for other sectors change in response to crop production, the same mechanisms can be derived as a difference-in-differences equation for indirect revenue, which corresponds closely with our empirical estimation.[14]

3.2 Spillovers and Observable Margins of Adjustment

This section relates the mechanisms in Equation (2) to observable margins of adjustment of sectors. While Equation (2) is useful for understanding the mechanisms, we do not estimate structural parameters, such as Λ_j, because some channels are not directly observable in historical data. Instead, we use observable changes corresponding to each term in the equation to infer the relative contribution of underlying mechanisms.

For the local demand effect, direct estimation would require data on local output prices for each sector, some of which are unavailable in historical settings. Instead, we rely on observable

[14] Note that the corresponding indirect revenue is $(P_{\mathrm{jct}}\,\Theta_{\mathrm{jct}})^{\frac{1}{\gamma}} W_{jct}^{-\frac{1-\gamma}{\gamma}} (1-\gamma)^{\frac{1-\gamma}{\gamma}}$

data such as changes in population and wages to infer shifts in local demand. Increases in population and wages plausibly reflect outward shifts in local demand, which is a common approach in the local spillovers literature (Hornbeck and Keskin 2015; Allcott and Keniston 2018; Bleakley and Lin 2012).

For the agglomeration effect, we observe the number of producers and define it as the extensive margin in each sector. Equations (1) and (2) indicate that an increase in the number of producers is necessary for a positive agglomeration effect. However, an increase in the number of producers is not sufficient for a positive agglomeration effect because the agglomeration spillover coefficient may be zero. Thus, our empirical approach tests the necessary condition for a positive agglomeration effect by estimating the effect on the number of producers in each sector.

For the coagglomeration and input cost effects, we cannot measure these channels directly in the historical data. Of the four channels in Equation (2), there are only three channels through which crop production can generate positive spillovers to other sectors: local demand, agglomeration, and coagglomeration. Therefore, if we observe sustained increases in the total value of output without increases in local demand or the number of producers, we interpret the gains as evidence that positive coagglomeration effects are substantial enough to outweigh the negative input cost effect.

4 Data Description

This section describes the study area, data sources, and variables used in the analysis. Our study area focuses on counties overlying the High Plains Aquifer and the surrounding region within 150 kilometers of the aquifer border (Figure 1).[15] Extending our study area beyond this distance to include additional control counties would weaken the plausibility of the parallel trends assumption, potentially undermining the validity of our identification strategy. Appendix Figure A1 presents the results of a robustness test examining how changes in sample size affect our estimation results.

[15] We measure pairwise distances between the aquifer border and counties using the minimum Euclidean distance approach, which identifies the shortest straight-line distance between polygons. The range of counties within 150 km corresponds to approximately 259.3 km when measured from the county centroid instead.

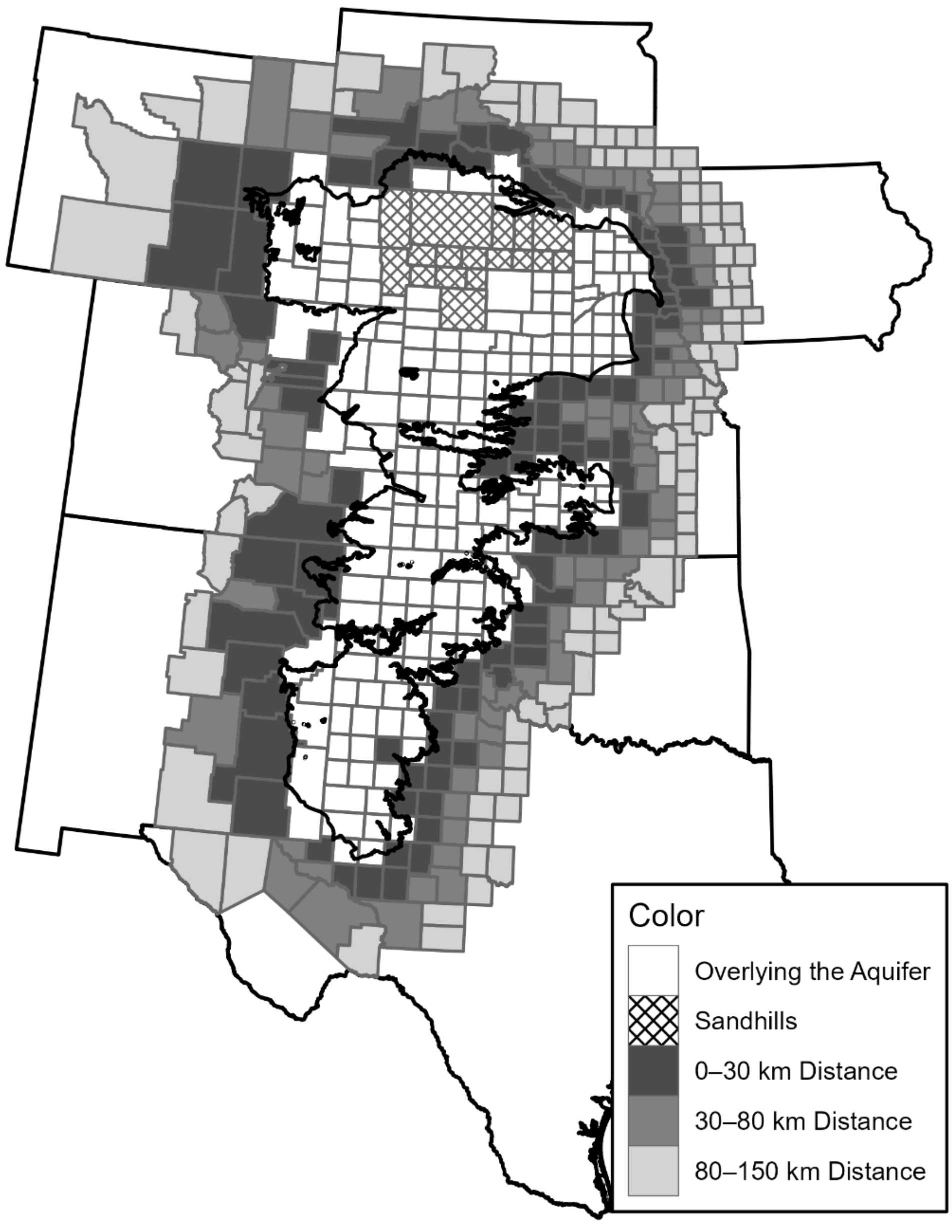

Figure 1. The High Plains Aquifer and counties within 150 km. This map shows counties overlying the High Plains Aquifer as well as those located within 150 km of its boundary. Counties are shaded by distance bands: 0–30 km (dark gray), 30–80 km (medium gray), and 80–150 km (light gray). The cross-hatched region marks the Nebraska Sandhills.

Our study area consists of 420 counties with the treatment group containing 184 counties located over the aquifer (Figure 1). We exclude 17 counties that are in the Nebraska Sandhills. Despite being located over the aquifer, these counties have extremely sandy soil that makes crop cultivation unfeasible (Perez-Quesada et al. 2024; Peterson et al. 2016). Thus, these counties have not experienced substantial crop production gains, so we exclude them from our study area.

There are 236 counties outside of the aquifer in our sample data. We use these counties to construct the control group and define two distance-based exposure groups, based on their distance from the High Plains Aquifer. The benefits of the aquifer could spillover beyond the aquifer boundary due to the tradability of crops and spatial equilibrium effects between regions. These spatial spillovers tend to diffuse into contiguous areas forming doughnut-shaped zones and attenuate with distance from the treatment region (Butts 2023). To account for this, we employ doughnut-shaped distance zones (Figure 1). Specifically, our distance zones delineate the two exposure groups and the control group: an inner exposure group within 0–30 kilometers, an outer exposure group within 30–80 kilometers, and a control group beyond 80 kilometers. The groups consist of 59, 74, and 103 counties, respectively.

We draw on historical county-level Census data over nearly a century (1920-2017) obtained from various sources. For data between 1920 and 1998, we exploit digitized versions of the quinquennial county-level Agricultural, Economic and Population Censuses (Gutmann 2005; M. Haines et al. 2018; M. R. Haines and Inter-university Consortium for Political and Social Research 2010). These datasets provide extensive historical information on agricultural production, land use, economic activity, and demographic changes across counties. For data between 2002 and 2017, we use data from the USDA National Agricultural Statistics Service (NASS) Quick Stats database and the U.S. Census Bureau's Economic Census. The NASS database provides quinquennial data on the agricultural sector, and The Economic Census collects detailed information on non-agricultural sectors.

We focus on six industrial sectors—crop, livestock, manufacturing, wholesale, retail, and service—and population. We analyze variables that capture the changes in total production, adjustments along the extensive and intensive margins, and sources of local demand shifts. For the livestock sector, we define the extensive margin as changes in the number of livestock farms and the intensive margin as changes in per-head value and the inventory of animals per farm. For non-farm sectors, the extensive margin refers to changes in the number of establishments, while the intensive margin captures changes in sales per establishment and employment. To capture local demand shifts, we use changes in total population and non-farm sector wages as key sources of local demand.

Table 1 provides summary statistics for our 37 dependent variables. For the crop sector, production refers to total bushels harvested and acres denotes the total area harvested. Yield per acre is calculated as bushels harvested divided by acres harvested. For the livestock sector, livestock product sales represent the total dollar value of livestock products sold. The number of cattle farms and hog farms corresponds to the number of farms reporting each type of livestock. Per-head value for cattle and hogs is calculated by dividing total sales by the number of animals sold. For the non-farm sectors, value added and total sales are taken from sectoral economic reports. The number of establishments captures the count of firms within each sector. Value added per establishment, sales per establishment, and employment per establishment are computed by dividing the respective totals by the number of establishments. As local demand related variables, we use total population and wages in non-farm sectors including manufacturing, wholesale, retail, and services.

The starting and ending years vary across variables due to differences in data availability (see columns 6 and 7 of Table 1). While some variables, such as harvested acres for wheat, have records dating back to 1920, others, like livestock product sales and non-farm sector metrics, begin in later years. Ideally, we would have several pre-treatment periods of data (before 1950), but some variables only have one pre-treatment period available.

The proportion of zero observations is highest for crops: 8.0% for wheat production and acres, 23.6–23.7% for sorghum production and acres, and 15.7–15.8% for corn production and acres (column 8 of Table 1). Although the prevalence of zeros may be a concern in log specifications, Appendix Figure A2 shows similar results when we use variables normalized by county acres as the dependent variables instead.[16] There are few zeros for other sectors.

Table 1. Summary statistics for dependent variables.

Variable	Mean	SD	Min	Max	Start Year	End Year	% zeros
Crop Sector							
Corn Production (mil. bu)	3.2	6.1	0.0	53.2	1920	2017	15.8
Sorghum Production (mil. bu)	0.6	1.3	0.0	15.8	1925	2017	23.7
Wheat Production (mil. bu)	1.2	1.7	0.0	17.1	1920	2017	8.0
Corn Acres (1,000 ac)	38.1	52.1	0.0	339.7	1920	2017	15.7
Sorghum Acres (1,000 ac)	13.5	25.3	0.0	267.4	1925	2017	23.6
Wheat Acres (1,000 ac)	51.6	66.4	0.0	554.7	1920	2017	8.0
Corn Yield per Acre (bu/ac)	62.6	53.8	0.5	230.7	1920	2017	0.0
Sorghum Yield per Acre (bu/ac)	35.6	24.9	0.9	151.5	1925	2017	0.0
Wheat Yield per Acre (bu/ac)	22.8	12.2	0.7	97.3	1920	2017	0.0
Livestock Sector							
Livestock Product Sales ($mil.)	101.4	169.1	0.0	3,254.1	1930	2017	0.7
Number of Cattle Farms	641.3	590.9	0.0	5,064.0	1930	2012	0.1
Number of Hog Farms	308.0	454.4	0.0	3,285.0	1935	2012	3.3
Cattle inventory per Farm	154.3	280.0	3.0	10,540.8	1930	2012	0.0

[16] This method uses shares as outcome variables and weights the regressions by county acres, which allows zero observations to remain in the sample. Hornbeck and Keskin (2014) also adopt this approach.

Hog inventory per Farm	227.3	1,642.9	1.0	86,011.4	1935	2012	0.0
Per-Head Value of Cattle ($)	1,370.3	396.3	450.6	2,713.4	1940	2017	0.0
Per-Head Value of Hog ($)	448.3	199.4	9.2	2,129.2	1940	2017	0.0
Manufacturing Sector							
Value Added ($mil.)	150.6	634.3	0.0	10,226.4	1920	1997	4.4
Number of Establishments	25.3	77.1	0.0	1,193.0	1920	1997	2.8
Value Added per Establishment ($mil.)	2.2	3.6	0.0	51.2	1920	1997	0.0
Employment per Establishment	20.7	23.3	0.3	238.7	1920	1997	0.0
Wholesale Sector							
Total Sales ($mil.)	444.9	2,213.6	0.0	44,538.5	1940	1997	1.5
Number of Establishments	46.2	124.8	0.0	2,067.0	1930	1997	1.6
Sales per Establishment ($mil.)	5.0	4.0	0.1	49.9	1940	1997	0.0
Employment per Establishment	5.0	3.3	0.2	54.9	1930	1997	0.0
Retail Sector							
Total Sales ($mil.)	269.7	828.0	0.0	13,638.7	1930	1997	0.1
Number of Establishments	213.2	408.2	0.0	5,530.0	1930	1997	0.1
Sales per Establishment ($mil.)	1.0	0.7	0.1	19.4	1930	1997	0.0
Employment per Establishment	3.7	2.4	0.1	18.1	1930	1997	0.0
Service Sector							
Total Sales ($mil.)	57.7	351.4	0.0	10,118.2	1940	1987	0.6
Number of Establishments	146.4	421.3	0.0	7,267.0	1940	1987	0.6
Sales per Establishment ($mil.)	201.7	190.1	7.0	1,719.5	1940	1987	0.0
Employment per Establishment	2.1	2.1	0.1	16.6	1940	1987	0.0
Local Demand Related Variables							
Total Population (thousands)	19.8	47.1	0.0	660.4	1920	1997	0.0
Wages in Manufacturing ($)	6.2	7.1	0.0	51.9	1920	1997	0.1
Wages in Wholesale ($)	7.7	7.1	0.3	51.1	1930	1997	0.0
Wages in Retail ($)	4.5	3.7	0.0	36.5	1930	1997	0.0
Wages in Service ($)	4.7	3.9	0.0	45.0	1940	1987	0.2

Note: This table reports the mean, standard deviation, minimum, and maximum of all dependent variables used in the analysis, along with their start and end years and proportion of zero observations. Dollar values are deflated to 2022 dollars.

5 Econometric Methods

Our regressions leverage the county-level panel data, measuring differences in outcomes between counties over and outside the High Plains Aquifer through the past century. Although the boundary of the Aquifer is plausibly exogenous, there could be spurious spatial correlations in the cross-section. Therefore, we utilize a Difference-in-Differences identification strategy that relies on the weaker assumption of parallel trends over and outside the Aquifer. We estimate the relative effect β^r and absolute effect β^a in the flavor of Allcott and Keniston (2018). The relative effect β^r captures the relative difference in outcomes between counties over and outside the aquifer. The relative effect β^r partially includes the spatial spillover effects. In contrast, the absolute effect β^a indicates the effect of access to the aquifer on the treated compared to a counterfactual scenario of not having access to the aquifer. We obviate the spatial spillovers to estimate the absolute effect β^a.

Consider a two-way fixed-effects Difference-in-Differences (DiD) approach to identify the relative effect β^r of crop production induced by the access to the aquifer. The treatment group consists of counties overlying the High Plains Aquifer, and the control group includes all counties located within 0–150 kilometers from the aquifer boundary. Let $\ln(y_{ct})$ denote the log of an outcome of interest in county c at time t, α_c represent county fixed effects, and γ_t represent year fixed effects. The regression equation is as follows:

$$\ln(y_{ct}) = \alpha_c + \gamma_t + \beta^r \, 1\{c \in G\} \times 1\{t \geq 1950\} + \varepsilon_{ct}, \tag{3}$$

where $1\{c \in G\}$ is an indicator for whether county c overlies the High Plains Aquifer, and $1\{t \geq 1950\}$ equals one after the treatment year of 1950—the interaction of these two terms indicates treatment status. The relative effect β^r represents how outcomes changed over the aquifer compared to those outside the aquifer but is not the average treatment effect on the treated due to the potential for spatial spillovers. If the crop spillovers affect nearby control counties in the same direction as the treated counties, the estimator of β^r will underestimate the true treatment effect. Conversely, if the crop spillovers affect control counties in the opposite direction, β^r will overestimate the treatment effect (Butts 2023).

As an alternative to equation (3), a spillover-robust difference-in-differences identifies the absolute effect of access to the aquifer (Drysdale and Hendricks 2018; Roth et al. 2023; Butts 2023; Allcott and Keniston 2018; Clarke 2017). Let $1\{c \in d\}$ denotes an indicator for county c being in a doughnut-shaped zone d surrounding the aquifer. Then the regression to estimate the absolute effect is

$$\ln(y_{ct}) = \alpha_c + \gamma_t + \beta^a\, 1\{c \in G\} \times 1\{t \geq 1950\} + \sum_{d \in D} \theta_d^{spill} 1\{c \in d\} \times 1\{t \geq 1950\} + \varepsilon_{ct} \qquad (4)$$

where the set of distance zones $D = \{(0, 30], (30, 80)\}$ defines the two doughnut-shaped exposure zones, representing areas of 0–30 km and 30–80 km from the aquifer boundary, respectively. Adding the interaction terms $\sum_{d \in D} 1\{c \in d\} \times 1\{t \geq 1950\}$ changes the underlying control group to those counties beyond the doughnut regions (i.e., 80-150 km from the aquifer) and explicitly estimates the spatial spillovers in nearby surrounding counties. The coefficient β^a indicates the local economic spillovers of access to the aquifer in counties over the aquifer relative to the counterfactual of no access to the aquifer. The coefficient θ_d^{spill} indicates the spatial spillovers on counties in the distance zone d.

The absolute effect β^a is identified under the assumption that the spatial spillovers do not reach a distance greater than the maximum distance in the set D (i.e., 80 km). The assumption of no spatial spillover is more likely to hold when we broaden the maximum distance zone in the set D. However, there is a tradeoff in setting the maximum distance. Extending the maximum distance in D too far could reduce the plausibility of the parallel trends assumption if the counties in the control group become less similar to those overlying the aquifer. In our setting, the coefficient (θ_d^{spill}) for the furthest distance zone (30-80 km) is insignificant, indicating no evidence of spatial spillovers to our control group (>80 km).

We extend our analysis of equation (4) by conducting event studies to examine how the economic spillovers vary over time. The regression equation follows:

$$\ln(y_{ct}) = \alpha_c + \gamma_t + \sum_{k \neq 1945} \beta_k^a 1\{c \in G\} \times 1\{k = t\} + \sum_{k \neq 1945} \sum_{d \in D} \theta_{d,k}^{spill} 1\{c \in d\} \times 1\{k = t\} + \varepsilon_{ct}, \qquad (5)$$

where the coefficients β_k^a allow for dynamic effects of access to the aquifer relative to the baseline year, $t = 1945$, allowing us to track producer adaptation. For some variables, data in 1945 were not available, so 1940 is used as the baseline instead.[17] The coefficients $\theta_{d,k}^{spill}$ capture the year-specific spatial spillovers on counties within each distance zone d. This specification allows us to investigate the timing and persistence of treatment and spatial spillovers.

In the estimation of the above equations the dependent variables are in logarithmic form and treatment is binary, so we calculate $exp(\hat{\beta}) - 1$ to represent the proportional change in the

[17] The reference year of 1940 is used as the baseline for per-head livestock values, population, and variables related to non-farm sectors, due to data unavailability in 1945.

outcome. We cluster standard errors by the county to allow for heteroskedasticity and autocorrelation of the error terms over time.

As an alternative to Equation (5), one might consider a two-stage approach in which a crop production variable serves as the dependent variable in the first stage. However, it is unclear which measure of crop production would be most appropriate, such as total value of crop production, irrigated acres, or corn production. Furthermore, estimating this two-stage model would imply a particular mechanism through which access to the aquifer affects non-crop sectors—but a variety of mechanisms are possible. Because any specific crop variable would reflect only a subset of these mechanisms, we instead adopt a reduced-form approach, Equation (5), to estimate the total effect of aquifer access on non-crop sector outcomes.

Another reduced form approach would exploit variation in pre-development saturated thickness rather than a binary indicator of access to the aquifer. A larger saturated thickness matters for non-crop outcomes through the resulting increase in crop production from greater irrigation potential, even if direct effects of the aquifer on non-crop sectors are minimal. This alternative approach would allow us to explore whether the effects vary with the intensity of groundwater access. However, defining the appropriate saturated thickness measure (e.g., average in the county, average of some surrounding counties, or something other than an average) is not straightforward. Therefore, we prefer the reduced form approach that reflects an average effect of access to the aquifer, while reporting the saturated thickness results in the supplementary material.[18]

We also assess other potential econometric concerns such as omitted variables that change over time differently by region. Soil characteristics are typically time-invariant, so we account for

[18] Appendix Table A2 provides a supplementary heterogeneity exercise using group-specific DiD estimates, where treated counties are divided into low, middle, and high groups based on pre-development saturated thickness. These estimates indicate that the effects on livestock product sales are larger in counties with greater saturated thickness.

soils by employing county-fixed effects. While climate may change over time, we define the control group as counties within the aquifer's 150 km boundary, so given this close range, it is reasonable to expect that both groups have experienced similar changes in climates.[19] This rationale is also employed in other research (Hornbeck and Keskin 2014, 2015).

As another concern, while our dataset provides sufficient coverage of key variables as described in Section 4, the pre-treatment period is shorter for some of the dependent variables (see Table 1), which limits our ability to test the parallel trend assumption. However, our dataset includes a sufficiently longer pre-treatment period for key variables—corn production, cattle inventory, value added in manufacturing, and population (Figure A3 in the Appendix). Figure A3 in the Appendix displays the trends for the counties grouped by the distance from the Aquifer. All variables display similar pre-treatment trajectories between counties overlying and outside the Aquifer, and crop production and cattle inventory begin to diverge notably from the 1970s (Panels A and B of Figure A3), while other variables show no substantial divergence (Panels C and D of Figure A3). These results provide evidence to support the parallel trends assumption.

We also consider specifications with state-by-year fixed effects or soil controls interacted with year fixed effects in Section 6.5.2, but we prefer the baseline specification because treated counties in some states lack valid within-state comparison counties, and the additional controls do not improve precision for non-crop outcomes.

[19] Some may raise concerns about omitted weather shocks. Our focus is on the non-crop economic sectors, which respond to long-term expectations rather than short-term weather shocks, so differences in year-to-year weather realizations are not likely to drive differential trends across groups.

6 Empirical Results

For ease of exposition, this section begins with the event-study results from Equation (5), followed by the average treatment effect estimates from Equations (3) and (4), including their spatial spillover specifications.

6.1 Crop Production, Population and Wages

Our first set of empirical results traces out how access to the aquifer impacts crop production. While the impact on crop production is a bit obvious, it is important to confirm because our interpretation and the following analysis on other dependent variables rely on the expansion of crop production as a channel of influence. We estimate Equation (5) using the dependent variables of production, harvested acres, and yield for corn, sorghum, and wheat. These results capture absolute changes by controlling for the spatial spillover. Figure 2 displays that access to the aquifer led to significant increases in corn production. Corn production started to increase sharply in the 1960s, driven by positive adjustments along both the extensive and intensive margins (Figure 2). In 1969, corn production was 346.4% larger due to access to the aquifer, driven by a 109.1% increase in corn acreage and a 115.8% increase in yield per acre. The impact on corn production peaks at a 987.6% increase in 1997.

The pre-trends for corn exhibit irregularities, likely attributed to the Dust Bowl, which caused substantial disruptions to crop production during the 1930s. Roth (2022) notes that there are economic settings in which non-monotonic differences emerge in the pre-treatment period, and that relying solely on statistical pre-trend tests can be misleading, recommending instead the use of context-specific economic knowledge to guide the interpretation of how parallel trends might plausibly be violated in a given context. Roth (2022) also warns that selective reporting based on pre-trend significance can distort both point estimates and confidence intervals for post-treatment effects. Therefore, having a few statistically significant pre-trend coefficients that are small should not be a source of major concern for our identification strategy. Importantly, the sharp and substantial changes after 1950 lend support to our identification strategy and suggest that the estimated effects are not primarily driven by pre-existing trends since irrigation is the most plausible explanation for such a dramatic shift in outcomes between the area overlying and outside the Aquifer. Furthermore, our results from pre- and post-treatment periods are in line with Hornbeck and Keskin (2014) for the variables that they include in their analysis.

In contrast, the changes in sorghum and wheat remain noisy throughout the study period, showing no clear changes. This reflects the fact that sorghum and wheat are generally more drought-tolerant and less dependent on water availability, so improvements in groundwater access have had a more limited effect on their production. As a result, it is difficult to conclude that access to the aquifer significantly impacts production of either crop.

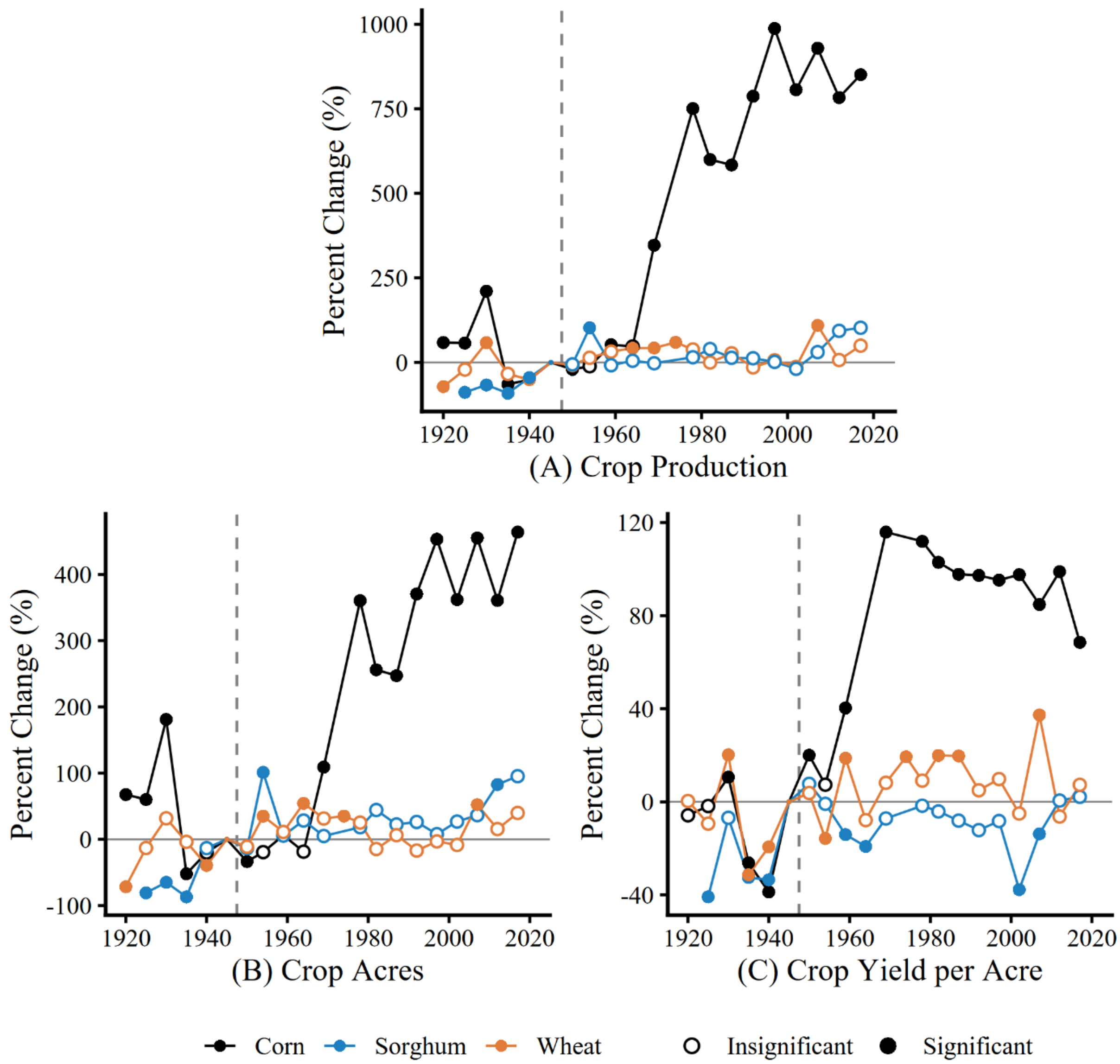

Figure 2. Event study results on the crop sector. (A) Percent changes in crop production. (B) Percent changes in crop acres. (C) Percent changes in yield per acre. The panels plot estimated coefficients from an event study regression of Equation (5) from 1920 to 2017. Filled circles indicate statistically significant coefficient estimates at the 5% level, while open circles indicate statistically insignificant estimates. Standard errors are clustered at the county level. Each line plots coefficients from a separate regression by crop: corn (black), sorghum (blue), and wheat (orange). The vertical dashed line marks the period just before the event year of 1950.

The second set of empirical results examines how crop production influences local population and wages. Figure 3A displays the year specific treatment effects on local population. The effect on population peaked around 1959 with a magnitude of 14.1% and then declined, becoming statistically insignificant thereafter. These intertemporal increases in local population align with results in Hornbeck and Keskin (2015), but we provide additional evidence on local demand increase in Figure 3B. Figure 3B paints a picture for effects on non-farm sectors wages. Wages in the wholesale sector significantly rose in 1959 by 11.6% and persisted through 1997, with the exception of insignificant effects in the 1964 through 1974 Censuses. Retail wages significantly increased in 1959 and 1964, although the effect did not persist over time, so it is hard to put too much weight on these results. Wages in the service and manufacturing sectors remained unaffected throughout the study period. Taken together, wages in non-farm sectors did not exhibit statistically significant declines; instead, population and wages rose together during the 1950s and 1960s, suggesting a short-run demand increase that retreated over time and an increase in wholesale wages that persisted over time.

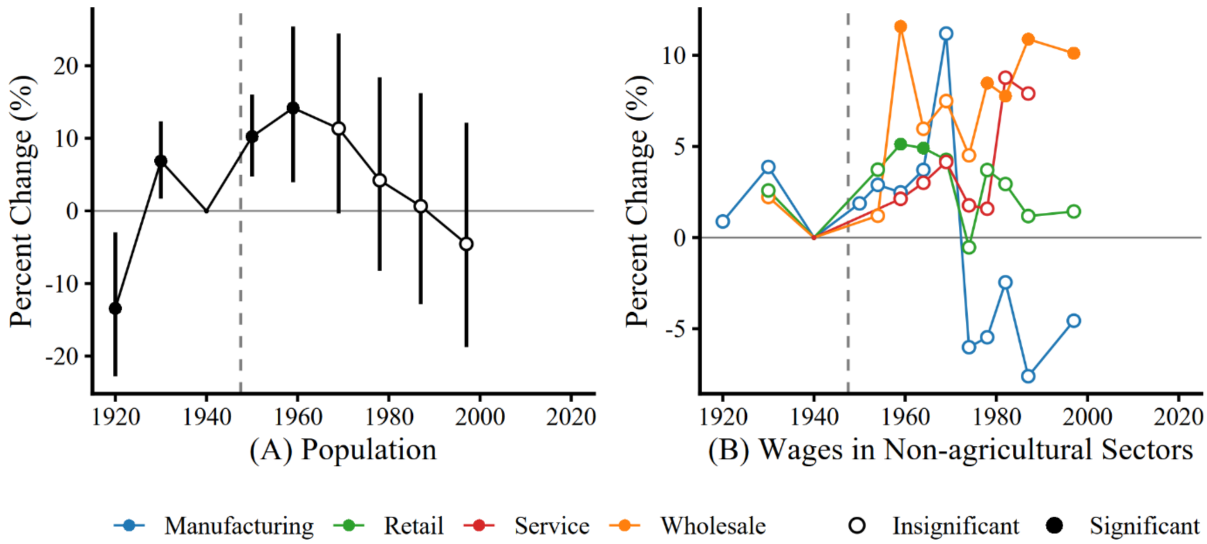

Figure 3. Event study results on the local demand. (A) Percent changes in population. (B) Percent changes in wages in non-agricultural sectors. The panels plot estimated coefficients from the event study regression of Equation (5) using data from 1920 to 1997. Filled circles indicate statistically significant coefficient estimates at the 5% level, while open circles indicate statistically insignificant estimates. Error bars in panel (A) represent 95% confidence intervals. Standard errors are clustered at the county level. Each line plots coefficients from a separate regression by sector. Colors in panel (B) represent different sectors: manufacturing (blue), retail (green), service (red), and wholesale (orange). The vertical dashed line marks the period just before the event year of 1950.

6.2 Livestock sector

Figure 4 shows that crop production generates strong spillovers on the livestock sector. The spillover effect on livestock product sales is significantly positive nearly every year after 1950, except for the initial two estimates in the 1950s (Figure 4A). By 1964, when corn production had begun to increase, the effect on livestock product sales significantly increased by 33.8%, and continued to increase over time, peaking at a 235.2% increase in 1997, remaining at a 196.9%

increase in 2017. In 2022 dollars, these estimates correspond to \$4.0 billion in 1969, \$23.6 billion in 1997, and \$29.9 billion in 2017.[20]

Second, for cattle producers, Figure 4C and Figure 4D show that they initially responded along the intensive margin through increases in per-head value, and later through expansions in inventory per farm. Per-head value increased immediately after 1950 and stabilized around a 25% increase beginning in the 1970s. In contrast, the estimated effects on inventory per farm were negative or statistically insignificant until the 1964 Census but gradually increased over time, surpassing over a 70% increase by the 1990s. These year-specific treatment effects paint a more nuanced picture of how cattle production adapted over time. The divergence between intensive and extensive margins in the 1950s is particularly telling. Rather than struggling to increase both margins in the early adaptation period, cattle farmers seemed to have prioritized increasing per-head value—likely reflecting a shift toward more intensive feeding practices of feeding cattle corn to finish weight rather than selling young calves. Over time, this was followed by gradual adjustments in the cattle inventory size.

In contrast, changes along the extensive margin are ambiguous (Figure 4B). Changes in the number of cattle farms are considerably noisier and do not provide clear evidence that crop production led to an increase in the number of cattle farmers. One limitation of the historical census data is that it does not distinguish between the types of livestock farms, in particular cow-calf grazing versus feedlot operations. As Opie, Miller and Archer (2018) note, many farms began feeding corn to their livestock during this period—a conversion from cow-calf to feedlot operations that is not explicitly distinguished in our data.

[20] We convert the estimates from Equation (5) into dollar values using the following equation: $y_{ck} - \frac{y_{ct}}{e^{\hat{\beta}_k^a}} = \frac{(e^{\hat{\beta}_k^a} - 1)}{e^{\hat{\beta}_k^a}} y_{ct}$, for $k \in \{1964, 1997, 2017\}$.

These results suggest the presence of substantial coagglomeration economics between crop and cattle farms. The theoretical predictions suggest three mechanisms could explain these positive effects: local demand, agglomeration, and coagglomeration. Livestock products are tradable goods and are not driven by local demand, so local demand cannot explain the increase. In addition, Figure 4B displays that cattle farms do not agglomerate within the region. Taken together, we interpret our findings as suggestive evidence that the observed positive spillovers on livestock sales are driven from the coagglomeration of crop and livestock producers.

This coagglomeration mechanism is reasonable given that livestock producers benefit from locally sourced feed, such as corn, which facilitates spatial linkages between the two sectors. Lower costs for feed by reducing transportation costs is especially important given the cost structure of feedlot production: feed-related expenses account for about 66–77% of total costs.[21] Appendix Tables A1 and A2 provide additional evidence that the livestock response is closely linked to groundwater-induced crop production. In counties overlying the aquifer, livestock product sales increased more in counties with a larger increase in corn production (Table A1). Treatment effects on livestock product sales are also larger in counties with higher predevelopment saturated thickness in 1930 (Table A2). These patterns are consistent with input-output linkages between corn and cattle, but they do not by themselves rule out other forms of coagglomeration, such as shared labor markets or knowledge spillovers. However, we view these channels as complementary rather than the primary explanation for coagglomeration in this setting. Shared labor markets may play some role, but the crop expansion was concentrated in mechanized field crops, for which hired labor represents a relatively small share of production costs. Furthermore,

[21] Historical USDA evidence indicates that feed accounted for about two-thirds to three-fourths of the cost of putting gain on cattle (Gustafson and Van Arsdall 1970). A 2014 beef-finishing budget shows that feed expenses account for 77% of total costs (Dhuyvetter and Tonsor 2014).

labor only accounts for about 1.9% of the cost of gain for beef cattle (Dhuyvetter and Tonsor 2014). Knowledge spillovers may also be a mechanism if new technologies or management practices in crop production affect the livestock sector. However, shared technologies and practices are relatively limited between these sectors, and it is not uncommon for farms in the region to specialize in either crop production or large-scale feedlots, so knowledge spillovers are unlikely to be a primary mechanism.

Dynamic complementarity may provide another source of crop-livestock coagglomeration. While crop production can support livestock production, expansion of feedlot activity may also reinforce crop production by improving the local crop price basis and reducing transportation, storage, and contracting costs for crop producers, as well as through manure application and related complementarities.[22] Even if such feedback exists, it does not invalidate our identification or reduced-form interpretation because access to the aquifer is plausibly exogenous and the initial shock operates through irrigation-induced crop production. Instead, the complementarity represents an additional source of coagglomeration incentives within agriculture.

One potential concern is that the estimated effects may simply reflect direct effects on the livestock sector rather than spillovers from crop production. We consider this possibility in Section 6.5.1. Another possible concern is that the results may be sensitive to additional controls. We discuss this issue in Section 6.5.2.

[22] The widespread adoption of chemical fertilizers in the 1940s (Entwisle et al. 2005) suggests that manure was unlikely to be a major driver of the coagglomeration effects.

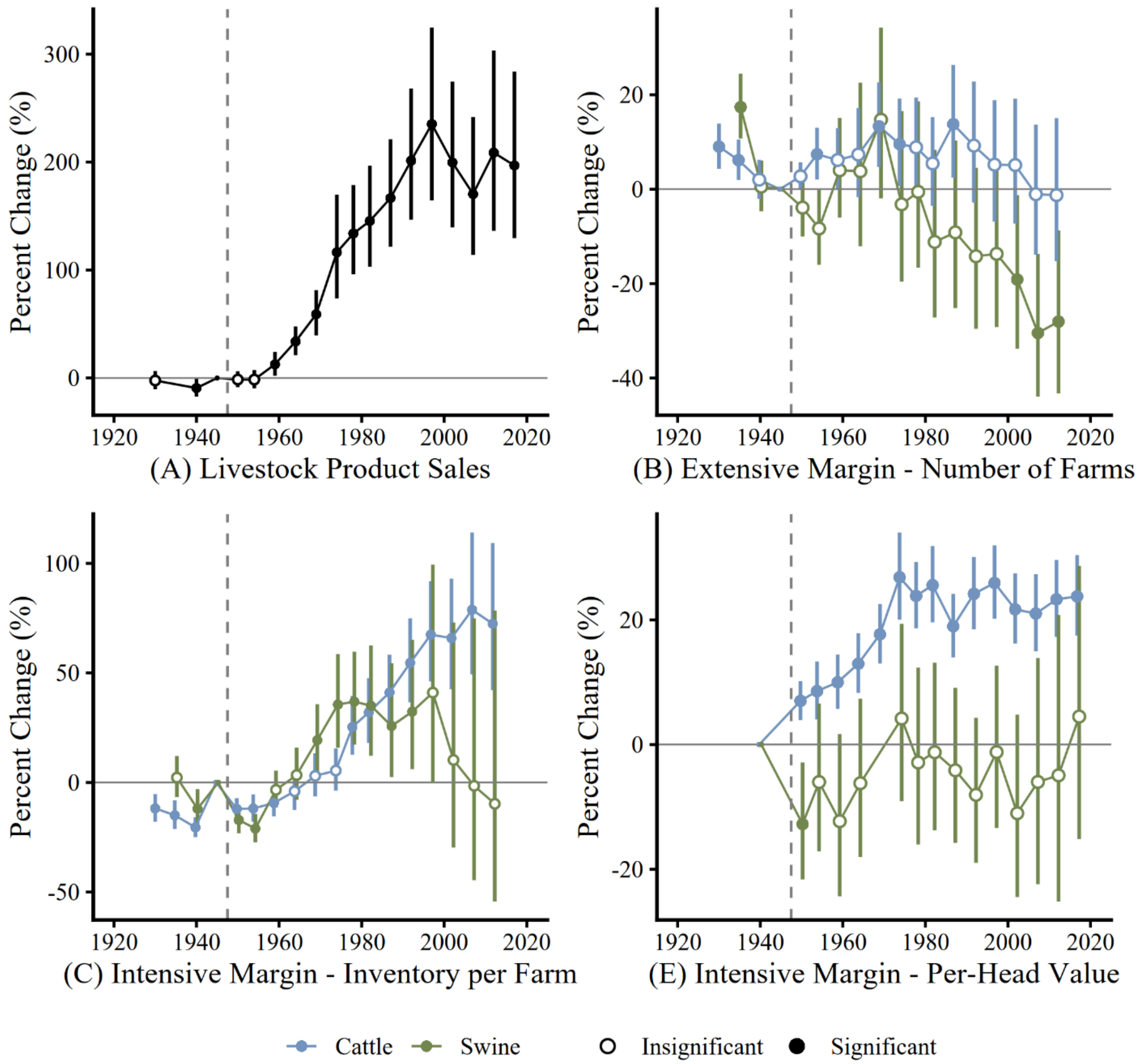

Figure 4. Event study results on the livestock sector. (A) Percent changes in total livestock product sales. (B) Percent changes in the number of livestock farms. (C) Percent changes in inventory per farm. (D) Percent changes in per-head value. The panels plot estimated coefficients from the event study regression of Equation (5) using data from 1930 to 2017. Filled circles indicate statistically significant coefficient estimates at the 5% level, while open circles indicate statistically insignificant estimates. Error bars represent 95% confidence intervals. Standard errors are clustered at the county level. Each line represents coefficients from a separate regression. In panels (B)–(D), colors distinguish livestock types: cattle (blue) and swine (green). The vertical dashed line marks the period just before the event year of 1950.

Crop production did not induce significant spillover effects on hog production in this region. Figure 4B shows the number of hog farms in the 2000s decreased significantly, but the pre-trend appears less clear, making it difficult to determine the true impact in the post-treatment period. The effects on hog inventory per farm are negative during the 1950s and 1960s, become significantly positive through the 1970s to 1990s, but return to statistically insignificant in the 2000s (Figure 4C). This ambiguity may be due to regulatory and environmental constraints, such as contamination management policies and zoning restrictions. For instance, Kansas' corporate farming laws impose zoning regulations and environmental compliance mandates, restricting hog production (Kansas Legislative Research Department 2013). Similarly, Nebraska's Corporate Farming Law places restrictions on hog inventory, capping hog farm expansion (Matthey and Royer 2001). These laws create barriers to expanding hog production in response to increased crop supply. More broadly, these results suggest that spillover patterns depend on local conditions, including regulation, climate, and the structure of agricultural production. In our setting, the strongest evidence points to a corn-cattle linkage, but the weaker results for hogs suggest that this pattern does not apply across all input-output linkages. For example, in California, where specialty crops are more central than corn, spillovers may operate through different agricultural linkages. Likewise, in wheat-oriented regions such as Montana and Washington, spillover patterns may differ from the corn-cattle linkage we document here.

6.3 Non-farm sector

Figure 5 illustrates event study results for the local spillovers of crop production on non-farm sectors. The results suggest consistently positive effects on wholesale sector outcomes, primarily along the intensive margins.[23] In the 1950s, the estimates indicate a sharp increase in wholesalers' sales. Throughout the entire study period, the treatment effects remain consistently positive and significant, and the upward trend continues with the peak estimate of a 68.1% increase in 1997. Figures 5C and 5D show that crop production increases revenue per establishment and employment in a pattern similar to that of total value added. In contrast, the spillover effects on the number of wholesale establishments are statistically insignificant across all census years. One likely reason for the increase in wholesale sales is that they may reflect increasing sales by firms providing inputs in crop and livestock production such as machinery, fertilizer, chemicals, and livestock supplies.

[23] To provide context for the magnitude of the wholesale sector, we calculate the sector's share of total GDP using county-level GDP by industry from the U.S. Bureau of Economic Analysis. The average share of wholesale trade in total GDP was approximately 5.00% in the treated region and 5.24% in the control region over 2001–2010.

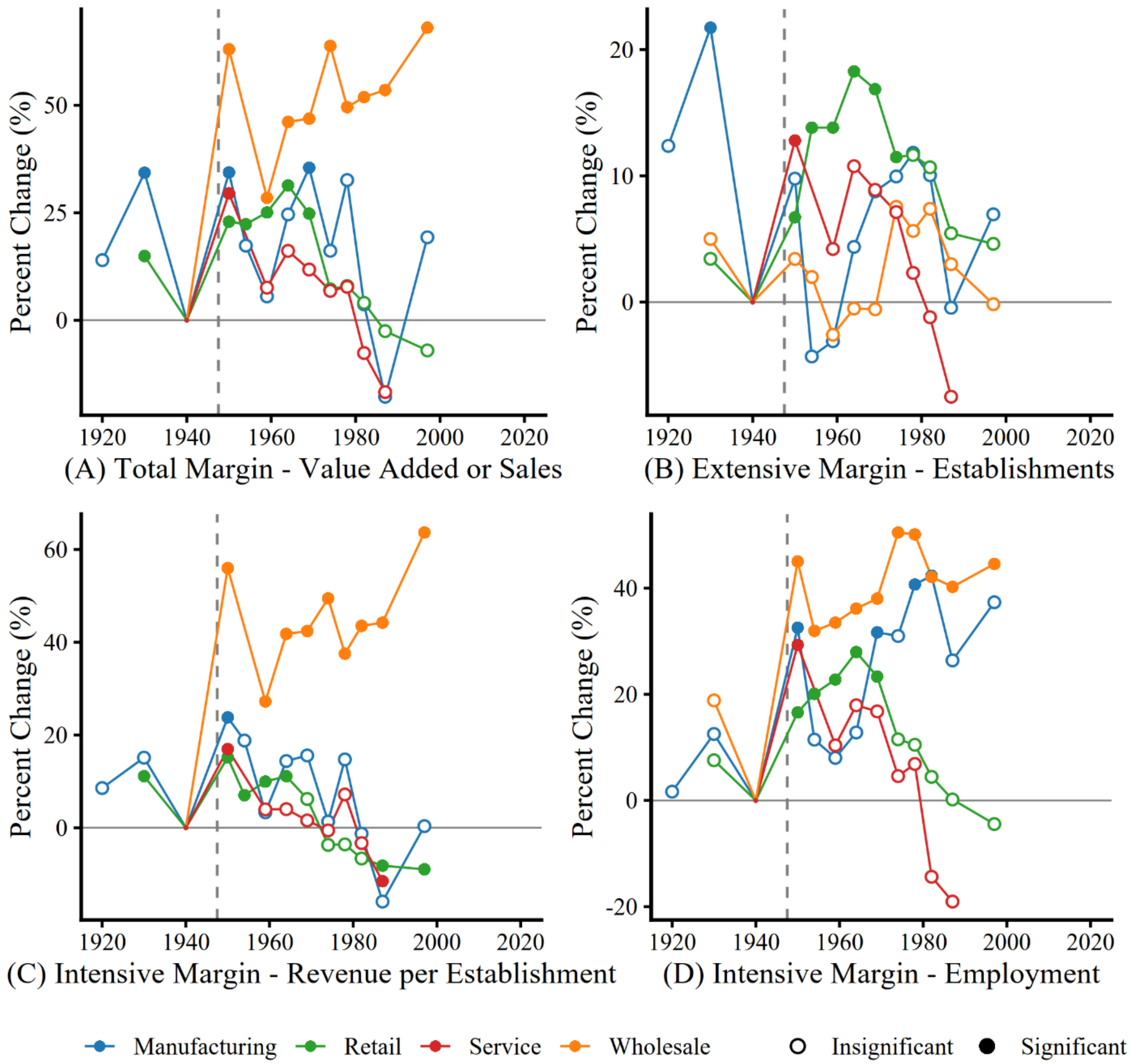

Figure 5. Event study results on the non-farm sector. (A) Percent changes in value added or sales. (B) Percent changes in number of establishments. (C) Percent changes in employment. (D) Percent changes in revenue per establishment. The panels plot estimated coefficients from the event study regression of Equation (5) using data from 1920 to 1997. Filled circles indicate statistically significant coefficient estimates at the 5% level, while open circles indicate statistically insignificant estimates. Standard errors are clustered at the county level. Each line plots coefficients from a separate regression by sector: manufacturing (blue), retail (green), service (red), and wholesale (orange). The vertical dashed line marks the period just before the event year of 1950.

Why does the wholesale sector adjust along with the intensive margins instead of the extensive margin? A sector tends to adjust with intensive margins if establishing a new establishment involves greater fixed and sunk costs than hiring workers into existing establishments (Allcott and Keniston 2018). Wholesale trade involves fixed infrastructure costs, established distribution networks, and relationship-based contracting, making entry barriers high even in the presence of increased demand. Instead of new firm formation, incumbent wholesalers capture the agricultural shock through expanded throughput and workforce increases.

On the other hand, the spillover effects on other non-farm sectors appear to be temporally limited. The manufacturing and service sectors exhibit a similar pattern in which total sales increase following 1950 but fail to persist in the long run. Retail sales increase after 1950 and remain elevated throughout the 1950s and 1960s, but lost significance in 1974. In contrast to wholesalers, retailers adjust more along the extensive margin than the intensive margin (Figure 5C). The increase in retail sales corresponds to the period where there was an increase in population (Figure 3A), which is intuitive since retail sales are driven by local demand. Overall, we find no evidence that the crop sector crowds out non-farm sectors and some evidence for a persistent increase in wholesale sales and short-run increases in retail sales.

6.4 Spatial Spillover

This section describes our results for spatial spillovers for the crop and livestock sectors. Model (1) of Table 2 reports estimates from Equation (3), which identifies the average relative treatment effect over the period. Models (2) and (3) report estimates from Equation (4), allowing for absolute effects while controlling for spatial spillovers through distance-based exposure zones but an average effect over time. Model (2) includes a single exposure zone of counties within 0–

30 km from the aquifer boundary with 30-150 km as the control, while Model (3) includes two zones, 0–30 km and 30–80 km, with 80-150 km as the control. To clarify, the two coefficients in the first row under Model (2) are from a single regression and the three coefficients under Model (3) are from a single regression.

Panel A in Table 2 displays the results for the crop sector. The estimated effects of aquifer access for counties directly overlying the aquifer are statistically significant and robust across all specifications. The magnitudes of the estimates remain largely stable when accounting for spatial spillovers, suggesting that our variation in aquifer access serves as a credible source of identification for changes in crop production. While Model (3) indicates statistically significant coefficients for spatial spillovers in corn yield and corn acreage, the coefficients are much smaller for the exposure areas than overlying the aquifer and event study results in the Appendix Figure A4 reveal that the corresponding pre-treatment trends are relatively noisy. Therefore, we do not interpret the statistically significant estimates in Model (3) of Panel A as providing strong evidence of spatial spillovers in crop outcomes.

Panel B in Table 2 reports the estimation results for the livestock sector. The estimated effects on livestock product sales over the aquifer are positive and statistically significant across Models (1)-(3). The coefficients are stable in magnitude—ranging from 105% to 114%—indicating robust effects even after controlling for spatial spillovers. In addition, intensive margin outcomes show robust patterns. In particular, results for the inventory per farm for cattle show an increase by approximately 41–42% across specifications, and the value per head of cattle shows a statistically significant increase, with estimates from 15.2% to 18.9%. Spatial spillover effects are found for the value per head of cattle in Model (3), though these results should be interpreted with caution. Specifically, the 0–30 km zone shows a statistically significant increase of 5.3% and the

30–80 km zone reveals a statistically significant increase of 6.3%. However, due to data limitations, we are unable to report pre-treatment coefficients for these zones and thus cannot confirm whether these effects are supported by stable pre-trends.

Results for the remaining dependent variables for the non-farm sectors and local demand are reported in the Supplementary Appendix Tables A3 and A4. The results support the conclusion of increases in wholesale sector activity over the aquifer and minimal impacts on other sectors. Out of the 156 estimates of spatial spillovers in the non-farm sectors, only 3 are statistically significant. The general lack of significant spatial spillovers in the farm and non-farm sectors provides empirical support for our identification strategy that the Stable Unit Treatment Value Assumption is likely to hold.

Table 2. Fixed effects difference in differences estimation.

Dependent Variable	(1)	(2)		(3)		
	Over the Aquifer	**Over the Aquifer**	**Within 0-30km**	**Over the Aquifer**	**Within 0-30km**	**Within 30-80km**
Panel A: Crop Sector						
Corn Production	2.963*** (0.556)	2.991*** (0.586)	0.033 (0.234)	2.618*** (0.62)	-0.063 (0.228)	-0.205 (0.157)
Sorghum Production	1.818*** (0.557)	2.037*** (0.635)	0.391 (0.56)	2.804*** (0.889)	0.742 (0.725)	0.615 (0.557)
Wheat Production	0.552*** (0.177)	0.555*** (0.189)	0.012 (0.209)	0.746*** (0.252)	0.135 (0.251)	0.304 (0.244)
Corn Acres	1.233*** (0.293)	1.209*** (0.307)	-0.052 (0.204)	0.897*** (0.311)	-0.186 (0.189)	-0.302** (0.138)
Sorghum Acres	1.377*** (0.425)	1.568*** (0.486)	0.408 (0.52)	2.13*** (0.665)	0.716 (0.655)	0.527 (0.483)
Wheat Acres	0.33** (0.15)	0.323** (0.161)	-0.025 (0.199)	0.507** (0.216)	0.111 (0.242)	0.35 (0.261)
Corn Yield	0.791*** (0.054)	0.829*** (0.057)	0.107* (0.063)	0.922*** (0.067)	0.163** (0.069)	0.125*** (0.046)
Sorghum Yield	0.177*** (0.038)	0.176*** (0.04)	-0.0004 (0.057)	0.19*** (0.047)	0.011 (0.06)	0.025 (0.051)
Wheat Yield	0.168*** (0.031)	0.178*** (0.033)	0.04 (0.046)	0.163*** (0.036)	0.026 (0.047)	-0.03 (0.04)
Panel B: Livestock Sector						
Livestock Product Sales	1.075*** (0.144)	1.052*** (0.147)	-0.051 (0.08)	1.142*** (0.172)	-0.009 (0.091)	0.108 (0.087)
Number of Cattle Farms	0.009 (0.03)	0.011 (0.032)	0.005 (0.05)	0.022 (0.039)	0.017 (0.055)	0.028 (0.045)
Number of Hog Farms	-0.11** (0.049)	-0.113** (0.052)	-0.015 (0.106)	-0.138** (0.061)	-0.042 (0.11)	-0.065 (0.086)
Cattle Inventory per farm	0.425*** (0.071)	0.41*** (0.072)	-0.049 (0.076)	0.414*** (0.078)	-0.047 (0.078)	0.006 (0.068)
Hog Inventory per farm	0.136* (0.073)	0.14* (0.079)	0.017 (0.103)	0.156 (0.095)	0.03 (0.114)	0.033 (0.107)
Value per Head of Cattle	0.152*** (0.017)	0.16*** (0.018)	0.032 (0.026)	0.189*** (0.022)	0.058** (0.029)	0.063** (0.025)
Value per Head of Hog	-0.066 (0.041)	-0.074* (0.041)	-0.042 (0.1)	-0.056 (0.051)	-0.023 (0.107)	0.047 (0.074)

Note: Robust standard errors are shown in parentheses. Treatment effects calculated as $e^{\beta} - 1$, based on coefficient β. Standard errors calculated using the Delta method. For each panel, estimation includes county and year fixed effects. ***p < 0.01; **p < 0.05; *p < 0.10.

6.5 *Additional Results to Support Identification and Robustness*

6.5.1 *Competing Explanations for the Increase in the Livestock Sector*

This section examines several competing explanations for the substantial increase in the livestock sector overlying the High Plains Aquifer. This matters for the interpretation of our identification strategy because our approach considers crop production as the primary channel through which treatment affects other sectors.

One potential explanation is that groundwater access after World War II could have directly affected the livestock sector by improving access to drinking water for livestock. However, stock water access is unlikely to explain most of the estimated treatment effect for several reasons. First, water for livestock was already available before World War II, so county fixed effects control for differences in access to groundwater for stock water, which is unlikely to change substantially after World War II. As mentioned in the historical discussion in Section 2, windmill-, steam-, and gasoline-powered pumps were available before World War II. Although they could irrigate only limited acreage for cropping, the pumps were sufficient for stock water (Meinzer 1911; Lugn and Wenzel 1938; Hutson 1898). Second, access to water for livestock is sufficient outside of the Aquifer in our study area. Appendix Figure A5 shows that the potential water available for stock water in counties outside the aquifer is as large, or larger than, stock water use over the aquifer. Therefore, differential access to drinking water does not provide a plausible explanation for our large treatment effects.[24] Finally, livestock only used 0.5% of groundwater in Kansas in 1987 (Buchanan and Buddemeier 1993), so livestock use a small portion of the groundwater.

[24] Appendix Figure A5 Figure A5maps USGS county-level water-use data for 2015 to compare actual livestock water use in treated counties with potential water availability for livestock drinking outside the aquifer. We assume that water used for irrigation outside the aquifer could alternatively be available for potential livestock water.

A second potential explanation for the increase in livestock is that access to groundwater may have increased irrigated pasture. However, this is also unlikely to explain the substantial increase in livestock sales. First, irrigated pasture acres account for less than 1% of total pasture overlying the aquifer from 1950 to 2012, making it implausible that irrigated pasture had a large impact on forage production. More specifically, Appendix Table A5 shows that the share of irrigated pasture acres was only 0.22-0.82% in treated counties from 1950 to 2012 and 0.24-0.75% in control counties over the same period. These small shares suggest that irrigated pasture is unlikely to be an important driver of our results. Second, Appendix Figure A6 shows counties overlying the aquifer sold 200% more fattened cattle in 1969 and 1,047% more in 1998 than control counties. This evidence suggests that corn-fed cattle feeding is likely a main driver of the results for increased livestock sales.

A third explanation for the increase in livestock sales is that favorable climatic conditions for cattle feeding explain the increase in feedlots in the region. But our analysis shows that areas just outside the aquifer with similar climatic conditions did not experience similar increases in livestock production, indicating that proximity to crop production was a key factor for the increase over the aquifer.

All of these factors—stock water, irrigated pasture, and climate—undoubtedly reflect benefits for the livestock sector to locate in the region. But, taken together, the evidence above indicates that these competing explanations are unlikely to account for a roughly 200% increase in livestock sales over the aquifer compared to nearby control counties.

6.5.2 *Additional Fixed Effects and Controls*

This section evaluates the robustness of the empirical results to additional fixed effects and controls. One potential concern is that counties may experience differential growth for reasons unrelated to the aquifer. State-by-year fixed effects control for differential changes over time by state. Intuitively, adding state-by-year fixed effects identifies the treatment effect using within-state, within-year comparisons. Appendix Tables A6-A9 show that the treatment effects for the main outcomes are larger with state-by-year fixed effects. However, we prefer our baseline specification because within-state comparisons are problematic in some important states. Figure 1 illustrates that none of the 31 counties in Iowa overlies the aquifer, while none of the 75 counties in Nebraska are in the 80-150 km control area, which indicates within-state comparisons are infeasible in these states.[25] Even if exposure counties (0-80 km) in Nebraska were included as control counties, most are located in the eastern part of the state and are not valid comparisons for a large portion of the state because of the east–west climate gradient.[26] Similarly, within-state comparisons in other states, such as Kansas and Texas, are also affected by climate gradients and controls only located in one or two directions. By contrast, restricting to county and year fixed effects allows comparisons to nearby control counties in other states and allows comparison counties to come from all directions, thereby averaging out geographic differences.

For adding controls, we consider time-interacted soil controls,[27] but we do not include them in the baseline specification because our main interest is in non-crop outcomes. If the parallel

[25] Iowa has no treated counties (0 of 31), with 4 counties in the 0–30 km group, 8 in the 30–80 km group, and 19 counties in 80-150km group. Nebraska has 66 treated counties, 5 counties in the 0–30 km group, and 4 counties in the 30–80 km group, and no counties in 80-150km group.

[26] Precipitation and humidity increase gradually from west to east in the Great Plains, making eastern counties systematically different from western counties.

[27] We consider NCCPIs and soil textures from SSURGO data as the controls. NCCPI refers to the National Commodity Crop Productivity Index. We add interactions between year indicators and county-level NCCPI measures for corn, soybeans, and cotton.

trends assumption holds, additional controls are unnecessary, but they may improve the precision of the estimates. More specifically, the controls could improve precision if they explain substantial variation in the outcome, but they could reduce precision if they are more correlated with the variation in the treatment than in the outcome. Appendix Figure A7 shows that the estimated effects on population and extensive-margin variables are similar to the baseline results. Figure A7 also indicates that the controls improve precision for crop outcomes but reduce precision for non-crop outcomes. This occurs because these controls absorb more variation in treatment than variation in the non-crop outcomes. We therefore do not include these controls in the baseline specification because our main interest is in non-crop outcomes, and coefficient estimates are similar with or without the additional controls.

7 Discussion And Conclusion

Increases in crop productivity are widely known to generate national and global economic gains, but the extent of localized economic spillovers remains a subject of debate. While theoretical channels offer competing predictions, we provide empirical evidence on the local economic spillovers of crop production to non-crop sectors. To do so, we exploit plausibly exogenous variation in groundwater access induced by post-WWII irrigation technologies across counties overlying and outside the High Plains Aquifer. Our estimates capture the absolute local spillover effects in both the short and long run, explicitly accounting for potential spatial spillovers into control counties.

To account for soil texture, we include interactions between year indicators and county-level shares of clay, sandy clay, sandy clay loam, sandy loam, loamy sand, sand, silty clay, clay loam, silty clay loam, loam, silt loam, and silt.

A key insight from our analysis is that we document that livestock production increased significantly over the past half century in the region due to the increase in crop production from the aquifer. On one hand, it might appear obvious that livestock feeding increased in the region because corn is a major input in livestock production. On the other hand, corn can be readily traded across counties, so livestock production did not necessarily need to increase within the region. Local increases in land values and wages from the expansion in crop production could even crowd out livestock production. However, we find that livestock sales increased substantially after the 1960s, peaking at a 235% increase compared to counties outside the aquifer in 1997. This growth appears to be driven by increases in both cattle inventories and per-head values, while the number of cattle farms did not exhibit a corresponding increase. Given that both local demand for livestock products and agglomeration among cattle producers appear limited, coagglomeration with crop producers—facilitated by input-output linkages—provides the most plausible explanation for the substantial expansion in cattle production over the aquifer.

In the broader economy, we find that wholesalers' total sales, employment and revenue per establishment increase in the short- and long-run, but we do not find a significant effect on the number of establishments. The increase in wholesale sector activity likely represents the sales of inputs and outputs within the agricultural supply chain. Wholesalers over the aquifer appear to adjust along the intensive margin, rather than through the entry of new firms, likely because establishing a new wholesale business entails larger fixed and sunk costs than expanding within existing firms. Meanwhile, there is evidence of increases in retailer and service sector activity in the short run, but the effects dissipate over time. The retail and service sectors are driven by demand from the local population, while wholesale is likely driven by demand from local agricultural production. We find no evidence of significant increases in the manufacturing sector, even though meatpacking plants that located in the region are included within the manufacturing sector. The reason is that manufacturing produces tradable goods that are not reliant on local demand. Meatpacking, while important, represents a fairly small share of activity within the region as a whole. Thus, our results indicate that crop production has minimal spillovers in sectors not within the agricultural supply chain.

There are four important caveats to our results. First, while our results show a large spillover to sectors within the agricultural supply chain, from crop to livestock, different local contexts may lead to different specific spillovers. For example, the spillover may occur for different types of livestock or for non-livestock sectors in the agricultural supply chain in different regions. Second, it is possible that the economic response to gaining access to groundwater is not symmetric to the response to losing it. In particular, expanded access to groundwater may trigger local investment and production increases, but the loss of access may not lead to equivalent contractions, especially if producers have already adjusted their operations or invested in infrastructure. Empirical literature on portage (Bleakley and Lin 2012) and the Tennessee Valley Authority (Kline and Moretti 2014) indeed find that local economies often continue to grow even after losing their initial incentives due to path dependence and agglomeration economics. Third, our analysis is based on aggregated livestock data, which do not distinguish between cow-calf and feedlot operations. While a transition from cow-calf to feedlot systems took place during the study period, such compositional shifts within the livestock sector are not separately observed. As such, some extensive margin adjustments in the type of livestock farms are not observable in our data. Lastly, while we view input-output linkages between corn and cattle as the primary mechanism, we cannot completely rule out complementary roles for shared labor markets, specialized services, or knowledge spillovers.

8 References

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Supplementary Appendix A

This section shows how Equation (2) follows from the conceptual setting. Taking logs of the indirect profit function $\pi_{jct} = \left(P_{\mathrm{jct}}\,\Theta_{\mathrm{jct}}\right)^{\frac{1}{\gamma}} W_{jct}^{-\frac{1-\gamma}{\gamma}}\,\gamma(1-\gamma)^{\frac{1-\gamma}{\gamma}}$ and plugging in $\ln(\Theta_{jct})$ from Equation (1) gives

$$\pi_{jct} = \frac{1}{\gamma}\,p_{jct} + \frac{1}{\gamma}\left(\rho_{jt} + \rho_{jc} + \Lambda_j f_{jct} + \varsigma_j v_{jct}\right) - \frac{1-\gamma}{\gamma} w_{jct} + k, \qquad (3)$$

where lowercase letters denote variables in natural logs and $k = \ln\gamma + \frac{1-\gamma}{\gamma}\ln(1-\gamma)$. Equation (3) shows that indirect profits increase with output prices, sector-time productivity, sector-county suitability, agglomeration, and coagglomeration, but decrease with input costs. Let $\Delta^c x_{jt} \equiv x_{jat} - x_{jbt}$ denote the difference across counties a and b in period t. Taking county differences in Equation (3) gives

$$\pi_{jat} - \pi_{jbt} = \Delta^c \pi_{jt} = \frac{1}{\gamma}\,\Delta^c p_{jt} + \frac{1}{\gamma}\Delta^c \rho_j + \frac{\Lambda_j}{\gamma}\Delta^c f_{jt} + \frac{\varsigma_j}{\gamma}\Delta^c v_{jt} - \frac{1-\gamma}{\gamma}\Delta^c w_{jt}, \qquad (4)$$

where the sector-time productivity ρ_{jt} and the constant term k difference out because it is common across counties. Evaluating Equation (4) at $t = 0$ gives

$$\Delta^c \pi_{j0} = \frac{1}{\gamma}\,\Delta^c p_{j0} + \frac{1}{\gamma}\Delta^c \rho_j + \frac{\Lambda_j}{\gamma}\Delta^c f_{j0} + \frac{\varsigma_j}{\gamma}\Delta^c v_{j0} - \frac{1-\gamma}{\gamma}\Delta^c w_{j0}. \qquad (5)$$

Subtracting Equation (5) from Equation (4) for $t > 0$ gives

$$\Delta^c\pi_{jt} - \Delta^c\pi_{j0} = \frac{1}{\gamma}(\Delta^c p_{jt} - \Delta^c p_{j0}) + \frac{\Lambda_j}{\gamma}(\Delta^c f_{jt} - \Delta^c f_{j0}) + \frac{\varsigma_j}{\gamma}(\Delta^c v_{jt} - \Delta^c v_{j0}) - \frac{1-\gamma}{\gamma}(\Delta^c w_{jt} - \Delta^c w_{j0}), \quad (6)$$

where the regional difference in sector-county suitability $\Delta^c\rho_j$ differences out. Equation (6) shows that changes in indirect profits can be decomposed into changes in local demand, agglomeration, coagglomeration, and input costs. This equation is sufficient for our purposes, but for ease of exposition we assume that the two counties are symmetric at $t = 0$, so cross-county differences at period $t = 0$ are set to zero. Under this assumption, Equation (6) becomes

$$\Delta^c\pi_{jt} - \Delta^c\pi_{j0} = \frac{1}{\gamma}\Delta^c p_{jt} + \frac{\Lambda_j}{\gamma}\Delta^c f_{jt} + \frac{\varsigma_j}{\gamma}\Delta^c v_{jt} - \frac{1-\gamma}{\gamma}\Delta^c w_{jt}. \quad (7)$$

which corresponds to Equation (2) in the text.

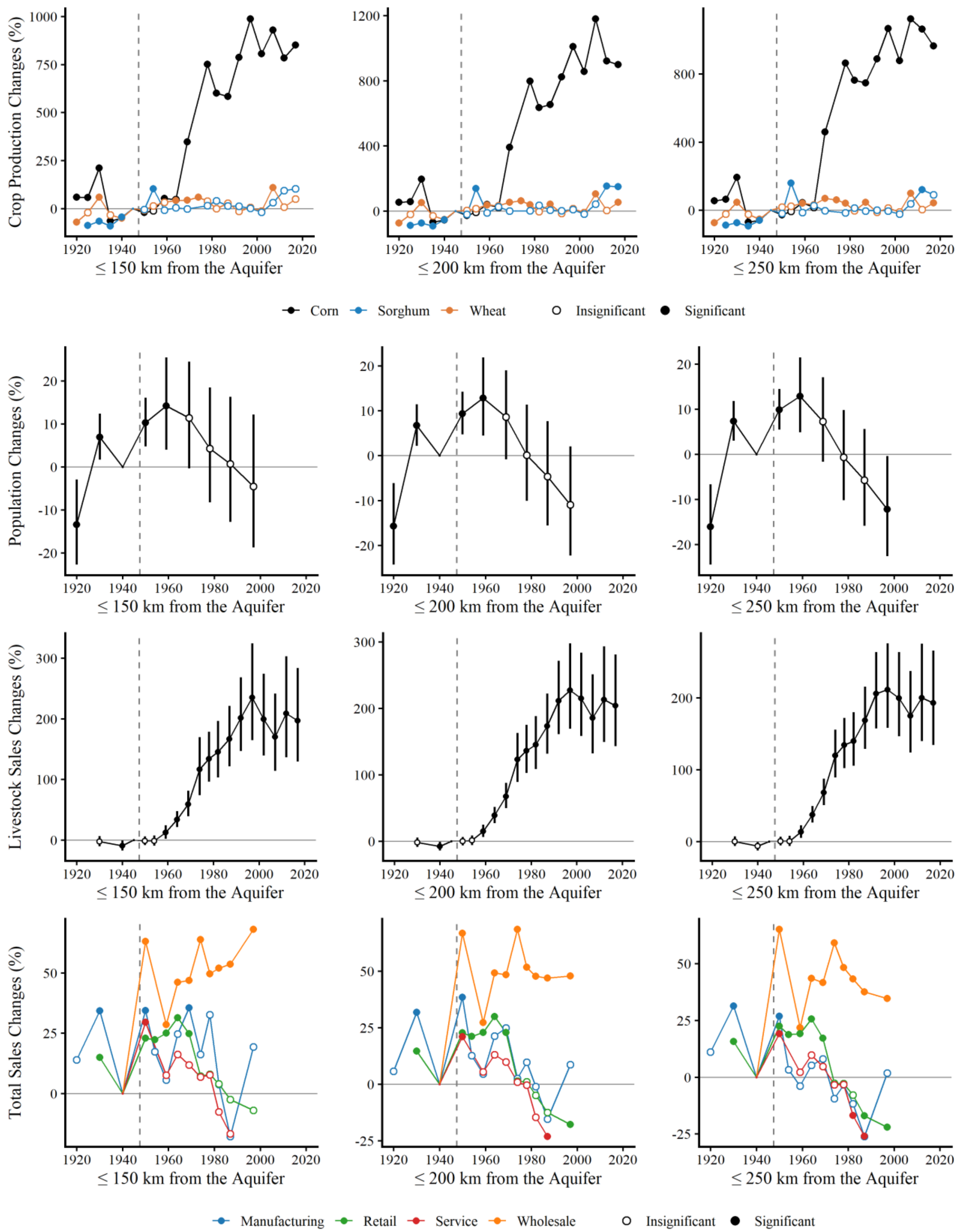

Figure A1. Robustness to alternative study area sizes. Each column presents event study estimates from Equation (5) using counties within 150 km, 200 km, and 250 km of the High Plains Aquifer boundary, respectively. Each row corresponds to a different outcome variable: percent changes in crop production, population, livestock sales, and total sales in non-agricultural sectors. Each line represents coefficient estimates from a separate regression. Error bars show 95% confidence intervals. Filled circles indicate statistically significant coefficients at the 5% level, while open circles indicate statistically insignificant estimates. Standard errors are clustered at the county level. The vertical dashed line marks the period just before the event year of 1950.

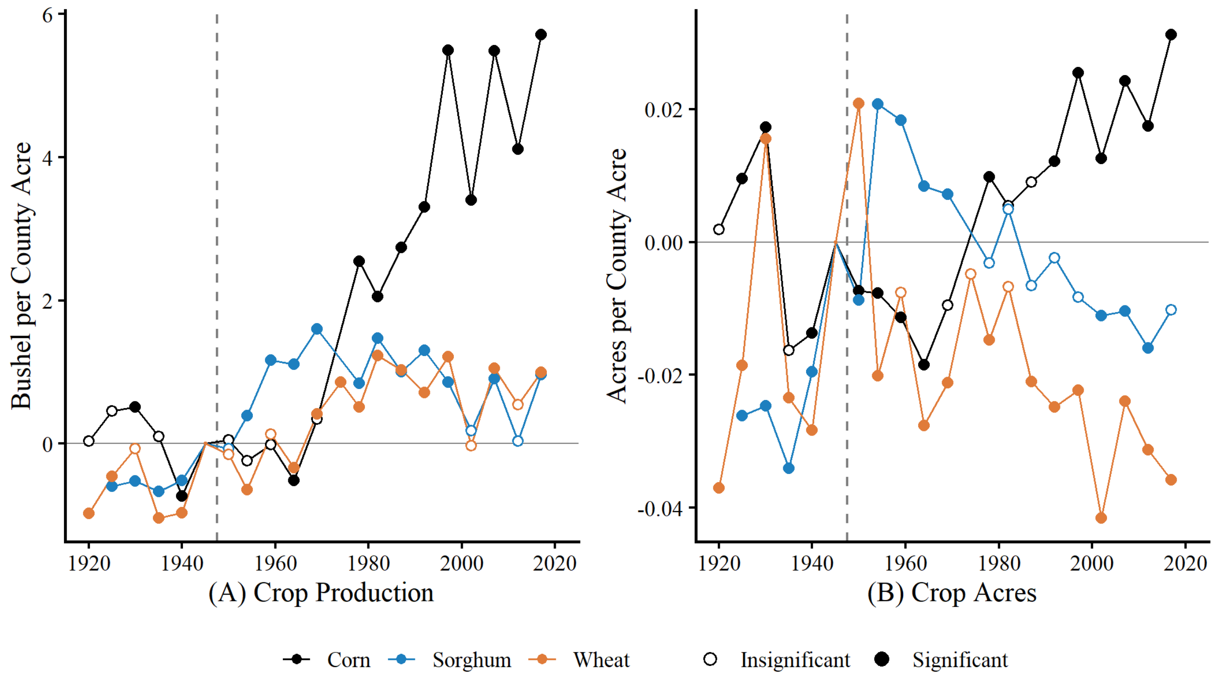

Figure A2. Event-study estimates for crop production and crop acres per county acre. (A) Crop production (bushels per county acre). (B) Crop acres (acres per county acre). This figure presents year-specific estimates from the following specification: $y_{ct} = \alpha_c + \gamma_t + \sum_{k\neq 1945} \beta_k^a 1\{c \in G\} \times 1\{k = t\} + \sum_{k\neq 1945} \sum_{d \in D} \theta_{d,k}^{spill} 1\{c \in d\} \times 1\{k = t\} + \varepsilon_{ct}$. Each line plots estimates from a separate regression: corn (black), sorghum (blue), and wheat (orange). Regressions are weighted by county acreage. Filled circles indicate statistically significant coefficients at the 5% level, while open circles indicate statistically insignificant estimates. Standard errors are clustered at the county level. The vertical dashed line marks the year just before the event in 1950.

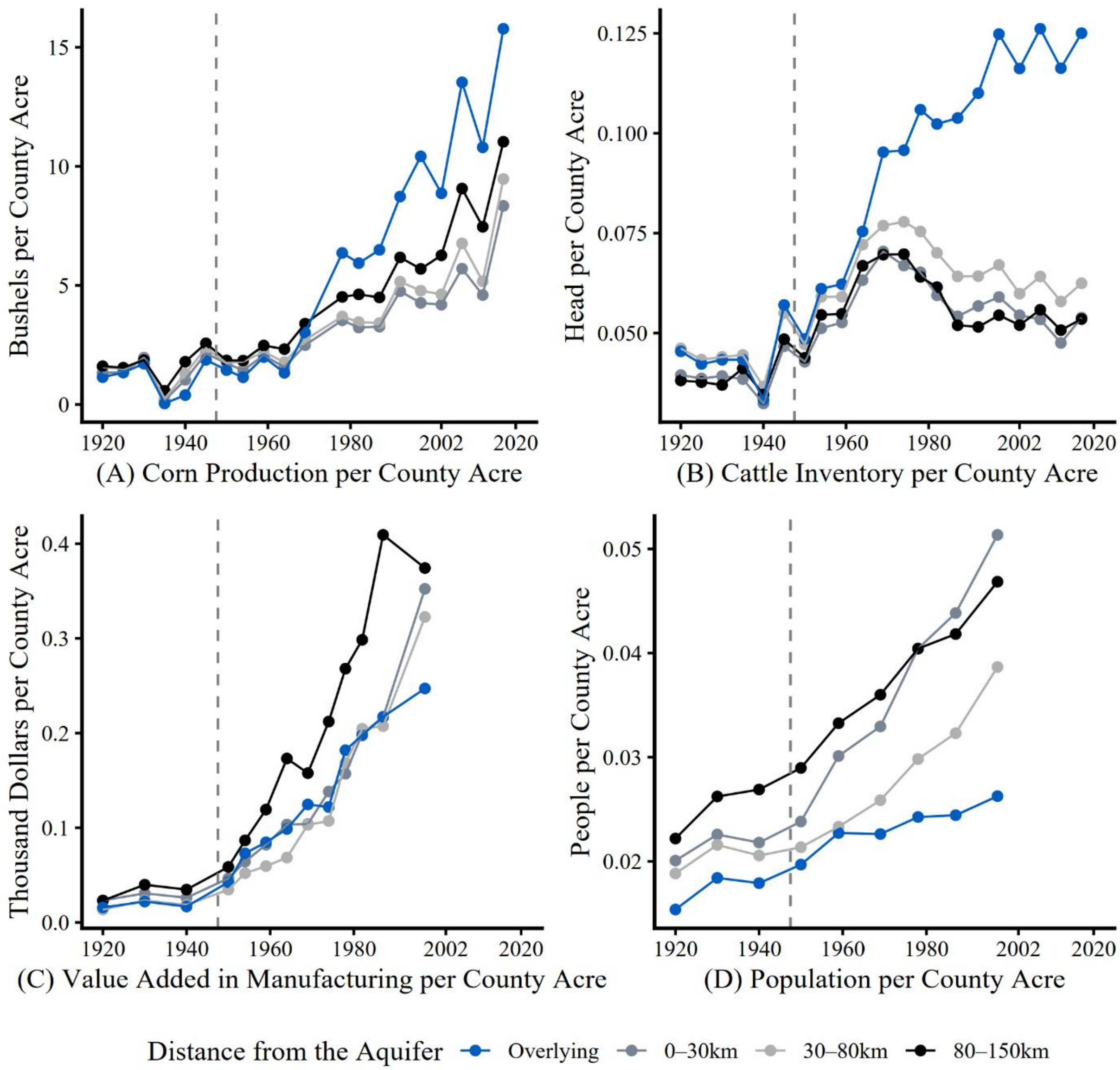

Figure A3. Parallel trends of counties overlying the High Plains Aquifer and surrounding counties. (A) Corn production per county acre. (B) Cattle inventory per county acre. (C) Value added in manufacturing per county acre. (D) Population per county acre. Points plot group-level totals divided by total county acres for each year from 1920 to 2017. Colors indicate distance bands from the aquifer: overlying the aquifer (blue), 0–30 km (dark gray), 30–80 km (light gray), and 80-150 km (black). The vertical dashed line marks the period just before the event year of 1950.

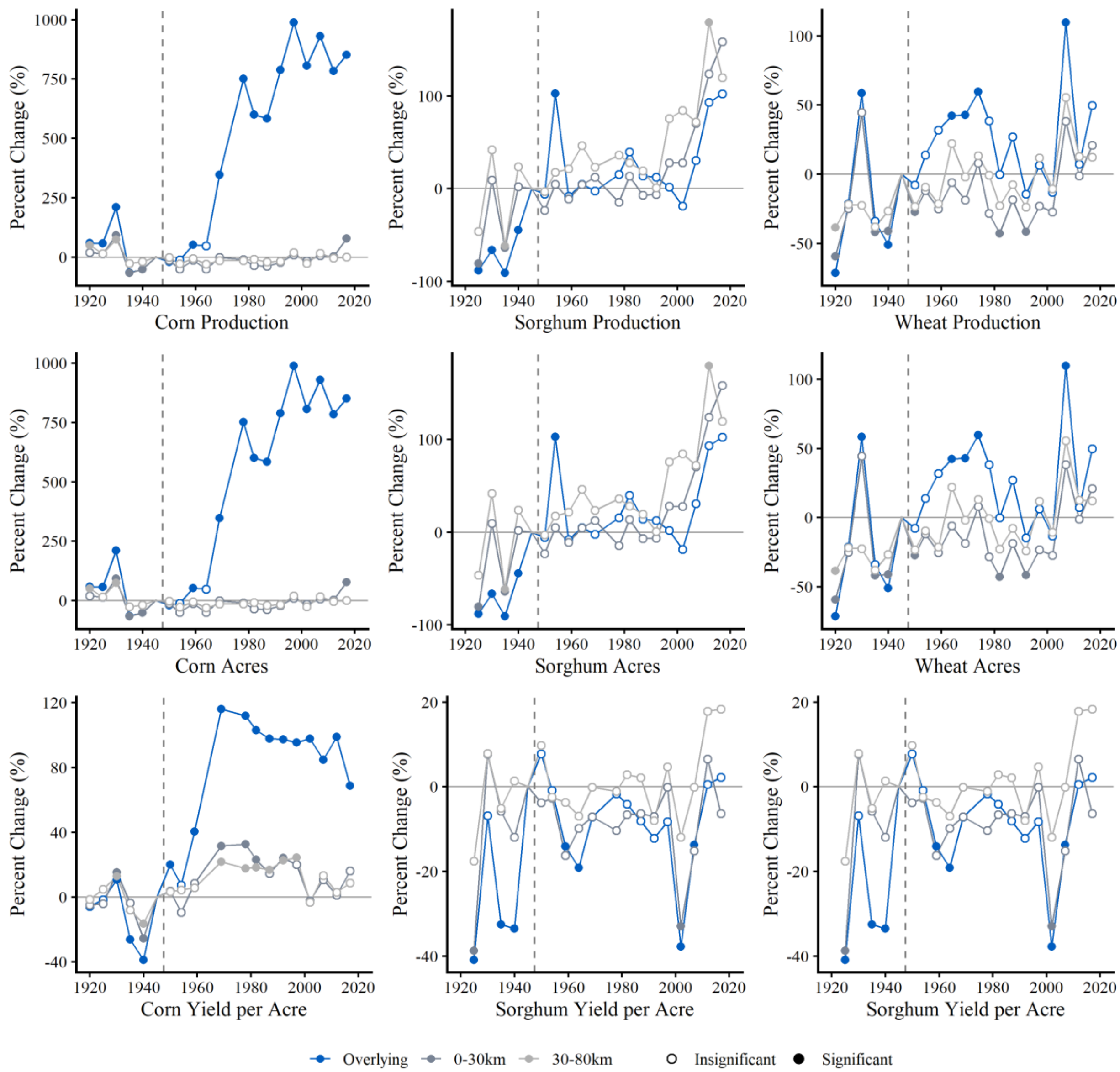

Figure A4. Event study results for counties overlying the High Plains Aquifer and the exposure zones. Each panel presents year-specific estimates in the crop sector from Equation (5), separately for counties overlying the High Plains Aquifer (blue), and those within 0–30 km (dark gray) and 30–80 km (light gray) from the aquifer boundary. Outcomes include total production (top row), harvested acres (middle row), and yield per acre (bottom row) for corn (left column), sorghum (center column), and wheat (right column). Filled circles indicate statistically significant coefficients at the 5% level, while open circles indicate statistically insignificant estimates. Standard errors are clustered at the county level. The vertical dashed line marks the year just before the event in 1950.

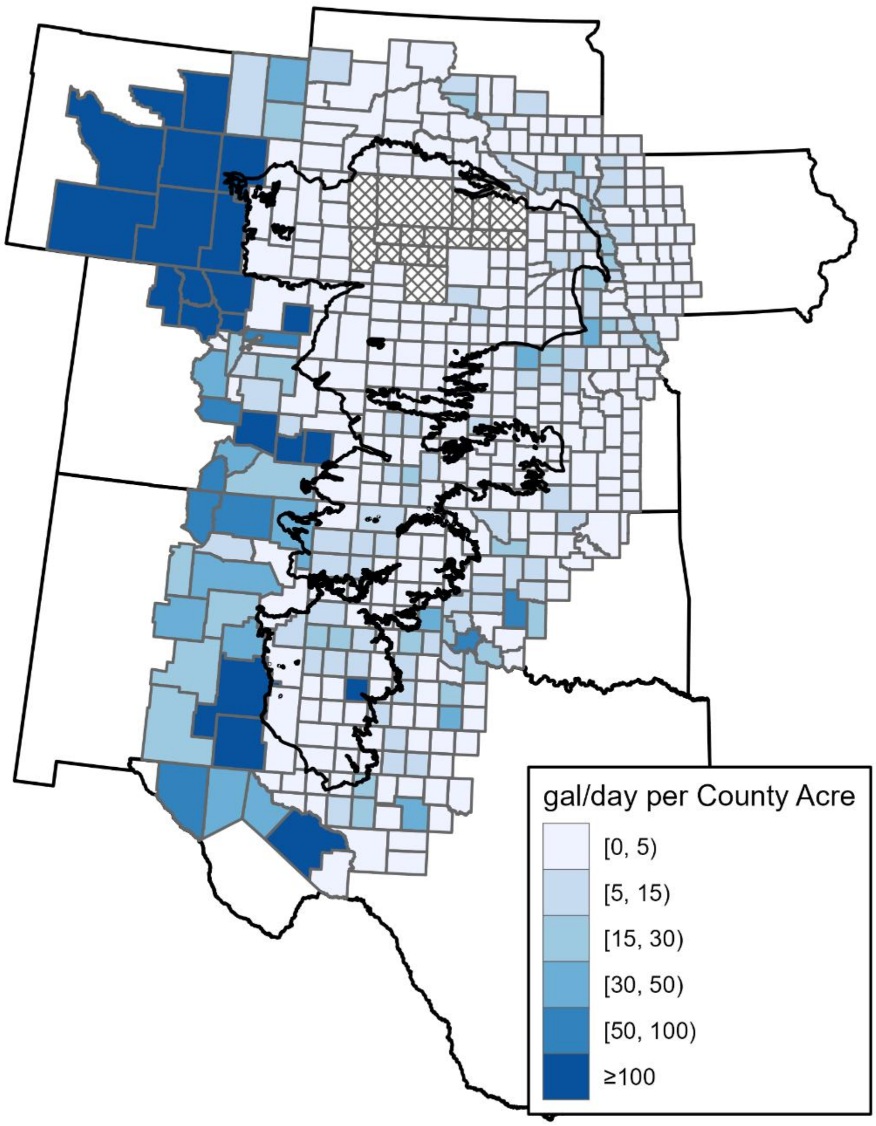

Figure A5. Comparison of livestock water over the aquifer to potential livestock water availability outside the aquifer. For counties overlying the High Plains aquifer, the map shows livestock freshwater withdrawals per county acre. For counties outside the aquifer, the map shows potential livestock water availability calculated as the sum of livestock freshwater withdrawals and irrigation withdrawals, under the assumption that irrigation withdrawals could alternatively be used for livestock. Water-use measures are based on USGS county-level data for 2015 and are expressed per county acre using county area from 1940. The cross-hatched region marks the Nebraska Sandhills.

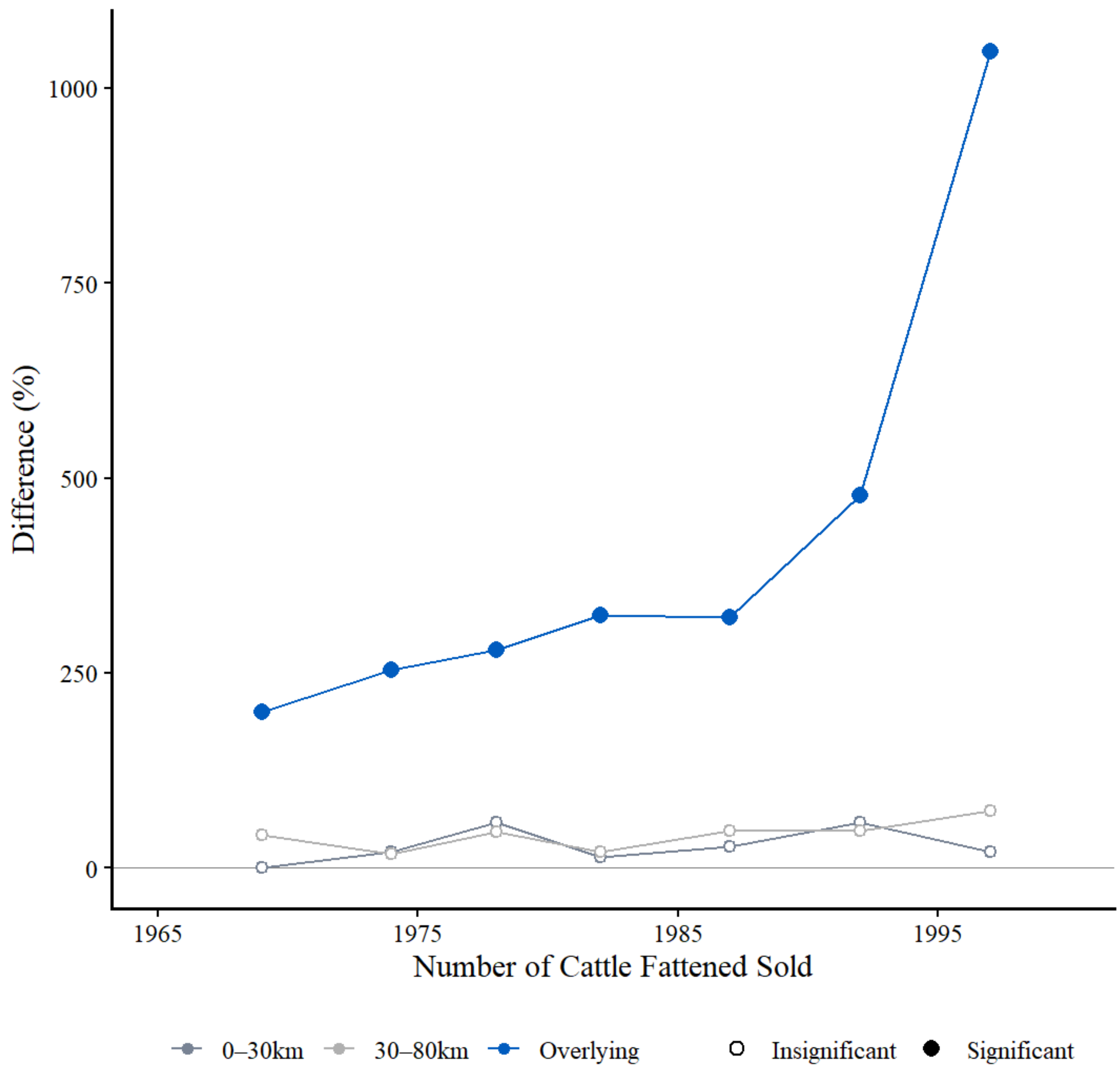

Figure A6. Cross-sectional regression results for counties overlying the High Plains Aquifer and the exposure zones. This figure presents year-specific estimates on the number of cattle fattened sold from the following specification: $\ln(y_{ct}) = \gamma_t + \sum_k \beta_k^a 1\{c \in G\} \times 1\{k = t\} + \sum_k \sum_{d \in D} \theta_{d,k}^{spill} 1\{c \in d\} \times 1\{k = t\} + \varepsilon_{ct}$. Lines show the cross-sectional differences relative to the baseline group for counties overlying the High Plains Aquifer (blue), counties located 0–30 km from the aquifer boundary (dark gray), and counties located 30–80 km from the aquifer boundary (light gray). The baseline group consists of counties located 80–150 km from the aquifer boundary. Filled circles indicate statistically significant coefficients at the 5% level, while open circles indicate statistically insignificant estimates. Standard errors are clustered at the county level.

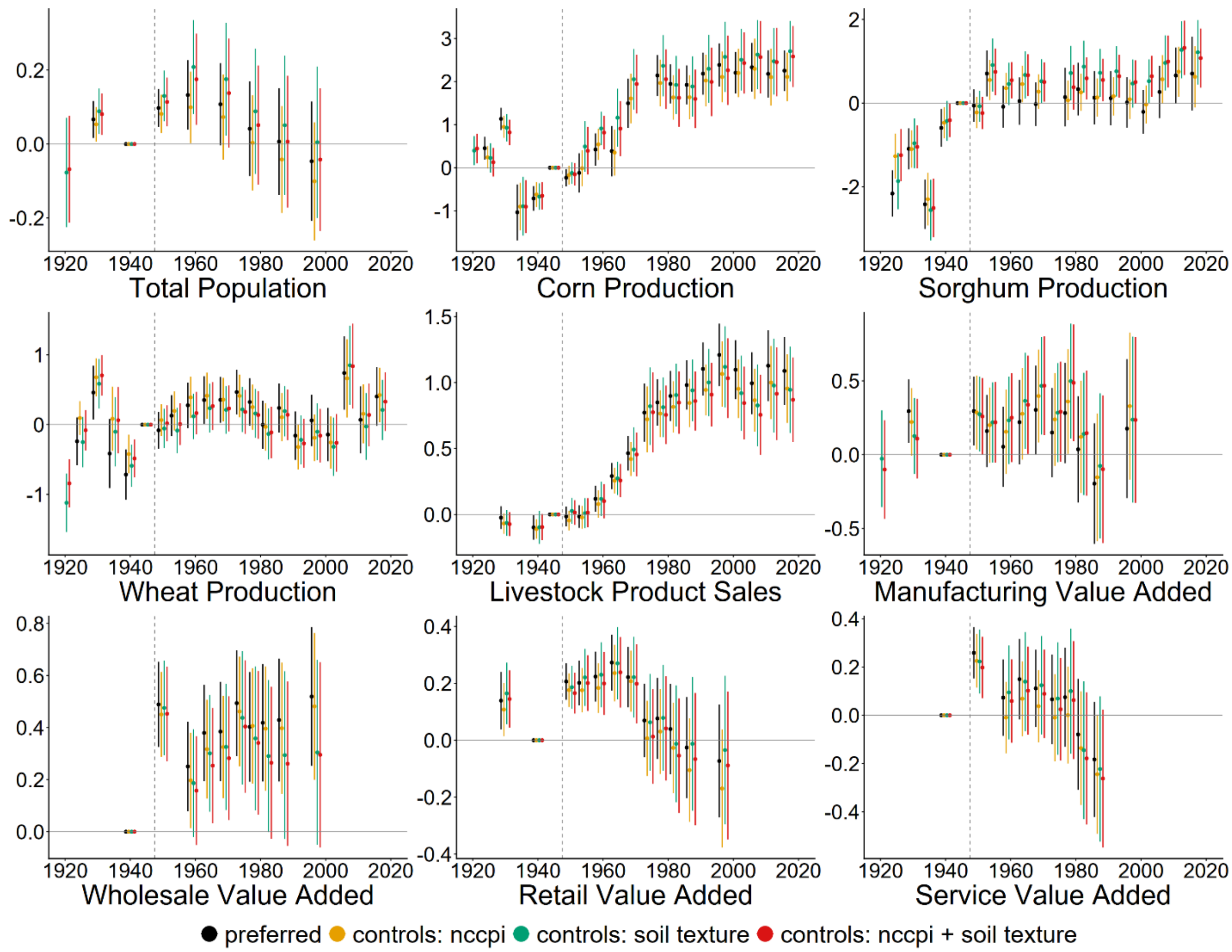

Figure A7. Robustness of event-study estimates to additional controls. Each panel presents year-specific estimates for the indicated outcome under four specifications: preferred specification with no additional controls (black), additional controls for NCCPI (gold), additional controls for soil texture (green), and additional controls for both NCCPI and soil texture (red). The NCCPI controls are constructed by interacting year indicators with county-level NCCPI measures for corn, soybeans, and cotton. The soil-texture controls are constructed by interacting year indicators with county-level shares of clay, sandy clay, sandy clay loam, sandy loam, loamy sand, sand, silty clay, clay loam, silty clay loam, loam, silt loam, and silt. Standard errors are clustered at the county level. Error bars show 95% confidence intervals. The vertical dashed line marks the year just before the event in 1950.

Table A1. Two-way fixed effects estimates of livestock product sales regressed on crop production over the aquifer

	Dependent Variable: Livestock Product Sales		
	(1)	**(2)**	**(3)**
Log Corn Production	0.1763***	0.1742***	0.1828***
	(0.0170)	(0.0168)	(0.0160)
Log Wheat Production		0.0231	0.0135
		(0.0281)	(0.0284)
Log Sorghum Production			-0.0886***
			(0.0139)
County FE	YES	YES	YES
Year FE	YES	YES	YES
R^2	0.79352	0.78832	0.80389
Observations	3,030	2,962	2,740

Note: This table reports estimates from regressions based on $\ln(y_{ct}) = \alpha_c + \gamma_t + \beta X_{ct} + \varepsilon_{ct}$, restricted to counties overlying the High Plains Aquifer. In columns (1)–(3), X_{ct} is log corn, wheat, and sorghum production, respectively. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A2. Group-specific DiD estimates by saturated thickness

	Dependent Variable: Livestock Product Sales
Low Saturated Thickness	0.3584***
	(0.0979)
Middle Saturated Thickness	0.6976***
	(0.0963)
High Saturated Thickness	1.127***
	(0.1122)
County FE	YES
Year FE	YES
$\boldsymbol{R^2}$	0.79209
Observations	7,498

Note: This table reports estimates from $\ln(y_{ct}) = \alpha_c + \gamma_t + \sum_{s\in(\text{low,mid,high})} \beta_s 1\{c \in G_s\} \times 1\{\text{t} \geq 1950\} + \varepsilon_{ct}$, where G_s is the set of counties overlying the High Plains Aquifer in saturated-thickness group s. Treated counties are grouped into low (below the 33rd percentile), middle (33rd–66th percentile), and high (above the 66th percentile) categories based on predevelopment saturated thickness in 1930. Standard errors clustered at the county level are reported in parentheses. *** p < 0.01; ** p < 0.05; * p < 0.10.

Table A3. Fixed effects difference-in-differences estimation for the non-farm sectors.

	(1)	(2)		(3)		
Dependent Variable	**Over the Aquifer**	**Over the Aquifer**	**Within 0-30km**	**Over the Aquifer**	**Within 0-30km**	**Within 30-80km**
Panel C: Manufacturing Sector						
Total Sales	0.032	0.015	-0.08	0.004	-0.09	-0.027
	(0.12)	(0.12)	(0.182)	(0.135)	(0.189)	(0.139)
Number of Establishments	-0.004	0.022	0.138	-0.042	0.067	-0.147**
	(0.057)	(0.059)	(0.133)	(0.063)	(0.13)	(0.066)
Revenue per Establishment	0.01	-0.032	-0.189*	-0.002	-0.164	0.08
	(0.081)	(0.081)	(0.106)	(0.095)	(0.116)	(0.119)
Employment	0.166	0.165	-0.004	0.181	0.009	0.034
	(0.108)	(0.11)	(0.158)	(0.127)	(0.169)	(0.116)
Panel D: Wholesale Sector						
Total Sales	0.449***	0.512***	0.231	0.516***	0.234	0.006
	(0.108)	(0.113)	(0.201)	(0.134)	(0.21)	(0.11)
Number of Establishments	-0.01	0.02	0.157	0	0.135	-0.046
	(0.061)	(0.065)	(0.137)	(0.076)	(0.143)	(0.092)
Revenue per Establishment	0.395***	0.408***	0.047	0.444***	0.074	0.063
	(0.082)	(0.086)	(0.11)	(0.101)	(0.118)	(0.093)
Employment	0.226***	0.278***	0.224	0.298***	0.243	0.037
	(0.084)	(0.089)	(0.188)	(0.108)	(0.199)	(0.112)
Panel E: Retail Sector						
Total Sales	0.046	0.077	0.151	0.055	0.127	-0.049
	(0.059)	(0.061)	(0.128)	(0.068)	(0.13)	(0.076)
Number of Establishments	0.086*	0.108**	0.099	0.094	0.085	-0.029
	(0.049)	(0.052)	(0.09)	(0.058)	(0.093)	(0.063)
Revenue per Establishment	-0.035*	-0.027	0.044	-0.036	0.033	-0.023
	(0.02)	(0.02)	(0.047)	(0.024)	(0.048)	(0.027)
Employment	0.073	0.103	0.139	0.089	0.125	-0.029
	(0.06)	(0.063)	(0.129)	(0.07)	(0.131)	(0.08)
Panel F: Service Sector						
Total Sales	0.036	0.054	0.086	0.062	0.094	0.017
	(0.071)	(0.075)	(0.164)	(0.087)	(0.171)	(0.116)
Number of Establishments	0.039	0.068	0.138	0.045	0.114	-0.05
	(0.057)	(0.06)	(0.136)	(0.069)	(0.138)	(0.084)
Revenue per Establishment	0.006	-0.008	-0.068	0.021	-0.041	0.071
	(0.03)	(0.031)	(0.051)	(0.036)	(0.054)	(0.052)
Employment	0.022	0.067	0.228	0.054	0.213	-0.029
	(0.08)	(0.087)	(0.177)	(0.097)	(0.182)	(0.118)

Note: Robust standard errors are shown in parentheses. Treatment effects calculated as $e^{\beta} - 1$, based on coefficient β. Standard errors calculated using the Delta method. For each panel, estimation includes county and year fixed effects. ***p < 0.01; **p < 0.05; *p < 0.10.

Table A4 Fixed effects difference-in-differences estimation for population and wages.

	(1)	**(2)**		**(3)**		
Dependent Variable	**On the Aquifer**	**On the Aquifer**	**Within 0-30km**	**On the Aquifer**	**Within 0-30km**	**Within 30-80km**
Panel G: Population and Wages						
Population	0.054	0.1	0.228	0.086	0.212	-0.031
	(0.068)	(0.071)	(0.155)	(0.078)	(0.157)	(0.083)
Manufacturing Wage	-0.004	-0.016	-0.057	-0.014	-0.056	0.004
	(0.024)	(0.025)	(0.037)	(0.029)	(0.04)	(0.032)
Wholesales Wage	0.07***	0.062**	-0.04	0.064**	-0.037	0.007
	(0.024)	(0.025)	(0.041)	(0.029)	(0.043)	(0.036)
Retail Wage	0.021**	0.02*	-0.007	0.017	-0.01	-0.007
	(0.01)	(0.011)	(0.017)	(0.013)	(0.018)	(0.013)
Service Wage	0.013	0.005	-0.039	0.042	-0.004	0.092**
	(0.03)	(0.032)	(0.033)	(0.036)	(0.037)	(0.042)

Note: Robust standard errors are shown in parentheses. Treatment effects calculated as $e^{\beta} - 1$, based on coefficient β. Standard errors calculated using the Delta method. For each panel, estimation includes county and year fixed effects. ***p < 0.01; **p < 0.05; *p < 0.10.

Table A5. Share of irrigated pasture overlying the aquifer and comparison counties

Year	Over the Aquifer	Within 0-30km	Within 30-80km	Within 80-150km
1950	0.22%	0.33%	0.26%	0.24%
1954	0.29%	0.32%	0.26%	0.26%
1964	0.47%	0.41%	0.34%	0.38%
1982	0.56%	0.31%	0.28%	0.32%
1987	0.65%	0.34%	0.30%	0.31%
1997	0.75%	0.67%	0.40%	0.42%
2007	0.82%	0.57%	0.38%	0.75%
2012	0.57%	0.58%	0.17%	0.42%

Note: This table reports the share of pastureland that was irrigated in counties overlying the High Plains Aquifer and in counties located 0-30km, 30-80km, and 80–150 km from the aquifer boundary. The share of irrigated pasture is calculated as irrigated pasture acreage divided by total pastureland acreage. The numerator includes only irrigated pasture acreage for 1950–1964, but includes irrigated pasture and other irrigated land for 1982–2012. Other irrigated land encompasses rangeland, failed cropland, cover crops, and other nonharvested irrigated land.

Table A6. Difference-in-differences estimates with two-way fixed effects: Crop sector.

	(1)		(2)		(3)	
Dependent Variable	**Coefficient**	R^2	**Coefficient**	R^2	**Coefficient**	R^2
Corn Production (n=7,040)	1.377*** (0.1403)	0.808	0.4907*** (0.1671)	0.712	1.212*** (0.1549)	0.869
Sorghum Production (n=6,096)	1.036*** (0.1975)	0.744	1.139*** (0.1630)	0.665	0.4824*** (0.1798)	0.854
Wheat Production (n=8,128)	0.4393*** (0.1139)	0.811	1.111*** (0.1599)	0.471	0.4861*** (0.1098)	0.867
Corn Acres (n=7,080)	0.8032*** (0.1312)	0.764	0.3454** (0.1549)	0.629	0.7007*** (0.1356)	0.839
Sorghum Acres (n=6,127)	0.8660*** (0.1786)	0.715	1.159*** (0.1512)	0.602	0.3642** (0.1658)	0.836
Wheat Acres (n=8,147)	0.2850** (0.1131)	0.826	1.152*** (0.1583)	0.453	0.2988*** (0.1015)	0.882
Corn Yield per Acre (n=6,667)	0.5829*** (0.0304)	0.885	0.1775*** (0.0320)	0.868	0.5331*** (0.0340)	0.912
Sorghum Yield per Acre (n=6,096)	0.1626*** (0.0323)	0.838	-0.0076 (0.0309)	0.795	0.1155*** (0.0315)	0.872
Wheat Yield per Acre (n=8,128)	0.1557*** (0.0269)	0.736	-0.0396** (0.0193)	0.736	0.1886*** (0.0254)	0.805
County FE	YES				YES	
Year FE	YES					
State-by-Year FE			YES		YES	

Note: This table presents estimates from Difference-in-Differences regressions based on Equation (3). Each column corresponds to a different fixed effects specification. Column (1) includes county and year fixed effects. Column (2) includes state-by-year fixed effects. Column (3) includes county and state-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. *** p < 0.01; ** p < 0.05; * p < 0.10.

Table A7. Difference-in-differences estimates with two-way fixed effects: Livestock sector.

	(1)		(2)		(3)	
Dependent Variable	**Coefficient**	R^2	**Coefficient**	R^2	**Coefficient**	R^2
Livestock Product Sales (n=7,498)	0.7299*** (0.0692)	0.787	0.7248*** (0.1018)	0.342	0.9113*** (0.0838)	0.807
Number of Cattle Farms (n=7,604)	0.0094 (0.0302)	0.941	-0.0277 (0.0812)	0.512	0.0243 (0.0368)	0.959
Number of Hog Farms (n=7,085)	-0.1168** (0.0552)	0.932	-0.0814 (0.0932)	0.753	-0.0705 (0.0579)	0.953
Cattle inventory per Farm (n=7,588)	0.3545*** (0.0495)	0.858	0.5382*** (0.0604)	0.649	0.4111*** (0.0619)	0.878
Hog inventory per Farm (n=6,603)	0.1276** (0.0640)	0.750	0.1924** (0.0745)	0.728	0.1493** (0.0657)	0.833
Per-Head Value of Cattle (n=6,438)	0.1414*** (0.0148)	0.857	0.1093*** (0.0160)	0.708	0.1129*** (0.0154)	0.884
Per-Head Value of Hog (n=5,206)	-0.0681 (0.0437)	0.502	0.0638*** (0.0219)	0.450	-0.0752 (0.0504)	0.562
County FE	YES				YES	
Year FE	YES					
State-by-Year FE			YES		YES	

Note: This table presents estimates from Difference-in-Differences regressions based on Equation (3). Each column corresponds to a different fixed effects specification. Column (1) includes county and year fixed effects. Column (2) includes state-by-year fixed effects. Column (3) includes county and state-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A8. Difference-in-differences estimates with two-way fixed effects: Non-farm sectors

	(1)		(2)		(3)	
Dependent Variable	**Coefficient**	R^2	**Coefficient**	R^2	**Coefficient**	R^2
Manufacturing Sector						
Value Added (n=3,319)	0.0312 (0.1161)	0.923	-0.6782** (0.2640)	0.456	-0.0394 (0.1469)	0.930
Number of Establishments (n=4,815)	-0.0039 (0.0575)	0.890	-0.4500*** (0.1349)	0.172	0.0262 (0.0719)	0.900
Value Added per Establishment (n=3,319)	0.0096 (0.0807)	0.906	-0.3874*** (0.1439)	0.655	-0.0924 (0.0968)	0.916
Employment per Establishment (n=3,375)	0.1537* (0.0927)	0.901	-0.2785*** (0.0919)	0.421	0.0147 (0.0828)	0.825
Wholesale Sector						
Total Sales (n=3,888)	0.3707*** (0.0743)	0.929	0.1726 (0.1580)	0.476	0.3986*** (0.0966)	0.934
Number of Establishments (n=5,013)	-0.0105 (0.0616)	0.867	0.0033 (0.1194)	0.116	0.0138 (0.0749)	0.887
Sales per Establishment (n=3,888)	0.3329*** (0.0591)	0.917	0.2057*** (0.0564)	0.821	0.3598*** (0.0629)	0.927
Employment per Establishment (n=4,585)	0.2036*** (0.0685)	0.892	0.0755 (0.0503)	0.496	0.2493*** (0.0544)	0.799
Retail Sector						
Total Sales (n=5,038)	0.0448 (0.0561)	0.942	-0.2466* (0.1412)	0.394	0.1001 (0.0724)	0.950
Number of Establishments (n=5,091)	0.0826* (0.0453)	0.931	-0.2994*** (0.1135)	0.190	0.1246** (0.0574)	0.940
Sales per Establishment (n=5,038)	-0.0360* (0.0212)	0.972	0.0597 (0.0372)	0.908	-0.0204 (0.0262)	0.975
Employment per Establishment (n=5,012)	0.0706 (0.0561)	0.927	-0.0022 (0.0414)	0.578	0.0116 (0.0300)	0.907
Service Sector						
Total Sales(n=3,736)	0.0358 (0.0687)	0.953	-0.3398** (0.1603)	0.500	0.1082 (0.0858)	0.958
Number of Establishments (n=3,799)	0.0382 (0.0547)	0.921	-0.2683** (0.1268)	0.156	0.1069 (0.0675)	0.934
Sales per Establishment (n=3,736)	0.0063 (0.0295)	0.967	-0.0465 (0.0444)	0.898	0.0118 (0.0359)	0.970
Employment per Establishment (n=3,639)	0.0222 (0.0782)	0.920	-0.0953* (0.0550)	0.627	0.0055 (0.0564)	0.866
County FE	YES				YES	
Year FE	YES					
State-by-Year FE			YES		YES	

Note: This table presents estimates from Difference-in-Differences regressions based on Equation (3). Each column corresponds to a different fixed effects specification. Column (1) includes county and year fixed effects. Column (2) includes state-by-year fixed effects. Column (3) includes county state-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table A9. Difference-in-differences estimates with two-way fixed effects: Local demand related variables.

	(1)		**(2)**		**(3)**	
Dependent Variable	**Coefficient**	R^2	**Coefficient**	R^2	**Coefficient**	R^2
Local Demand Related Variables						
Total Population (n=3,824)	0.0527 (0.0645)	0.861	-0.3144** (0.1268)	0.149	0.0974 (0.0818)	0.887
Wages in Manufacturing (n=3,045)	-0.0042 (0.0243)	0.966	-0.0011 (0.0298)	0.938	-0.0012 (0.0283)	0.969
Wages in Wholesale (n=4,175)	0.0681*** (0.0227)	0.961	0.0377* (0.0201)	0.941	0.0092 (0.0237)	0.966
Wages in Retail (n=4,589)	0.0210** (0.0102)	0.980	0.0342*** (0.0113)	0.969	0.0218* (0.0119)	0.982
Wages in Service (n=3,219)	0.0131 (0.0301)	0.945	-0.0211 (0.0226)	0.910	-0.0103 (0.0319)	0.948
County FE	YES				YES	
Year FE	YES					
State-by-Year FE			YES		YES	

Note: This table presents estimates from Difference-in-Differences regressions based on Equation (3). Each column corresponds to a different fixed effects specification. Column (1) includes county and year fixed effects. Column (2) includes state-by-year fixed effects. Column (3) includes county state-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.